

Battery intelligent temperature warning model with physically-informed attention residual networks

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HIGHLIGHTS

- Merged NDC models with differential networks for efficiency and interpretability.
- Introduced adaptive weighting for faster model training convergence.
- Used a Physics-Informed Neural Network with attention mechanisms to improve accuracy.
- Demonstrated model effectiveness on multiple datasets.
- Proposed a real-time training strategy for lifecycle management.

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ABSTRACT

The rapid development of electric vehicles demands improved thermal safety management of lithium-ion batteries. Traditional physical models require extensive offline parameter identification, struggling to balance computational efficiency and model fidelity, while data-driven methods, though precise, lack interpretability and require large datasets for varied conditions. To address these challenges, we propose the Physics-Informed Attention Residual Network (PIARN), which integrates an improved nonlinear dual-capacitor model and a thermal lumped model within a physics-guided recurrent neural network, enhancing both interpretability and generalizability. The residual attention network, comprising channel attention and time-series blocks, analyzes online measurements and hidden physical states to infer complex nonlinear dynamic responses, significantly improving accuracy. While a simplified physical model captures primary dynamics, the residual attention block corrects for missing nonlinear relationships. An adaptive weighting method accelerates network convergence by addressing voltage and temperature loss function magnitude discrepancy. Validation on three dynamic datasets demonstrates PIARN's ability to accurately predict battery voltage and temperature using sparse discharge data, showcasing strong generalization across varied conditions. Additionally, a cost-effective online iterative training framework is designed, enabling precise battery modeling and lifecycle tracking of aging and thermal status, with temperature prediction root mean square error as low as 0.1 °C and nearly 100 % accuracy in thermal warnings after multiple iterations. Thus, the novel PIARN model significantly enhance the accuracy of online temperature predictions and thermal warnings, thereby improving battery thermal management.

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Nomenclature**Abbreviations**

PIARN	Physics-Informed Attention Residual Network
EVs	Electric Vehicles
LIBs	Lithium-Ion Batteries
BTMS	Battery Thermal Management System
BMS	Battery Management System
CPCMs	Composite Phase Change Materials
NMC	LiNiMnCoO ₂
LFP	LiFePO ₄
DFN	Doyle Fuller Newman
SPM	Single Particle Model
ECM	Equivalent Circuit Model
LSTM	Long Short Term Memory
NARX	Nonlinear Autoregressive with Exogenous
NDC	Nonlinear Double-Capacitor
SPMT	Single Particle Model with Thermal
DST	Dynamic Stress Test
RNN	Recurrent Neural Networks
PINN	Physics-Informed Neural Network
PIRNN	PINN in the form of RNN architecture
AW	Adaptive Weighting
ECA	Efficient Channel Attention
Bi-GRU	Bidirectional Gated Recurrent Unit
NASA	National Aeronautics and Space Administration
OCV	Open Circuit Voltage
SOC	State of Charge
DCNN	Deep Convolutional Neural Networks
SE	Squeeze-and-Excitation
GRU	Gated Recurrent Unit
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R ²	coefficient of determination
Bat1–4	the four batteries of each group
ExpData	Experimental Dataset
SimData	Simulation Dataset
NASAData	NASA random Dataset
G6	Group 6
G7	Group 7
CC-CV	Constant-Current-Constant-Voltage
w./w.o.	with/without
ATTGRU	Attention-GRU
PIEN	Physical Information Embedded Network
PIAN	Physical Information Attention Network
PN	Physical Normalization
BN	Batch Normalization
LN	Layer Normalization
AI	Artificial Intelligence
TP	True Positives
TN	True Negatives
FP	False Positives
FN	False Negatives
NePCM	Nano-enhanced Phase Change Materials

Notations

R_b	the diffusion rate coefficient of lithium ions between the inner and surface layers
C_b	the amount of lithium ions in the inner regions
R_s	the diffusion rate coefficient of lithium ions between the surface layers and electrolytes
C_s	the amount of lithium ions in the surface regions
V_b	normalized virtual potentials of the inner layers
V_s	normalized virtual potentials of the surface layers

$q_{\max}$	the max total charge capacity
q_b	the actual charge capacity of the inner layers
q_s	the actual charge capacity of the surface layers
U	open circuit potential
R_0	ohmic resistance
R_1	liquid-phase diffusion resistance
C_1	liquid-phase diffusion capacity
q_1	the actual charge capacity of liquid-phase diffusion capacity
$h(\cdot)$	open-circuit potential function
C_{th}	heat capacity
R_{th}	thermal resistance
T_{th}	battery lumped temperature
t	time
I	current
V	terminal voltage
T_a	ambient temperature
$dU/dT(\cdot)$	the entropy function of the battery reaction
θ	the physical model parameter set
$\mathbf{H}$	the hidden state variable set
V_{lb}	the lower voltage bound
V_{ub}	the upper voltage bound
x	the state variable
x_{lb}	the lower bound of state variable
x_{ub}	the upper bound of state variable
C	the number of input channels
k	the size of the adaptive convolutional kernel
x_t	the input at the current timestep
h_t	the hidden state at the current timestep
z_t	the output of the update gate
r_t	the output of the reset gate
W_z	the weight matrix of the update gate
W_r	the weight matrix of the reset gate
W_h	the weight matrix of the new candidate hidden state
b_z	the biases of the update gate
b_r	the biases of the reset gate
b_h	the biases of the new candidate hidden state
S	the sequence length
V_{pred}	voltage prediction
V_{true}	voltage measurement
T_{pred}	temperature prediction
T_{true}	temperature measurement
w_1	adaptive weight for ℓ_V
w_2	adaptive weight for ℓ_T
T_r	the decay factor shaping temperature
D_{att}	the feature dimension of the ECA hidden layer
$S_{\max}$	the maximum sequence length
l	the learning rate
l_0	the initial learning rate
h	the prediction horizon
h_{th}	heat transfer coefficient

Symbols

β	the volume ratio between the inner and surface layers
α	maximum discharge rate
$\sigma(\cdot)$	sigmoid activation function
ℓ	loss function
ℓ_V	the loss term for voltage observation error
ℓ_T	the loss term for temperature observation error
ℓ_B	the loss term for boundary condition error
μ	dynamic shrinkage factor
ξ_1	the initial decay factor value
ξ_N	the final decay factor value

1. Introduction

Amid escalating concerns about global climate change and energy crises, the widespread adoption of Electric Vehicles (EVs) marks a pivotal shift toward more sustainable and environmentally friendly transportation [1,2]. Lithium-Ion Batteries (LIBs), characterized by their high operating voltage, substantial energy density, and extended cycle life, are predominantly utilized as the main power source for EVs. Nonetheless, batteries degradation and thermal safety issues persist as significant challenges in the development of LIBs and LIB-powered EVs [3,4]. Researches indicate that the performance and safety of power batteries are highly temperature-dependent. Operating under abnormal temperatures rapidly undermine LIBs' efficiency and stability [5]. Low temperatures significantly diminish battery capacity due to lithium plating during high-rate charge and discharge cycles, while high temperatures expedite side reactions and degradation, potentially triggering thermal runaway with severe risks [6]. Therefore, an appropriate Battery Thermal Management System (BTMS) is crucial to maintaining an efficient and robust system resistant to adverse internal and environmental temperature variations.

In recent years, there has been extensive research aimed at enhancing the thermal management level of energy storage systems in EVs. These studies have primarily focused on improving heat dissipation techniques, developing innovative materials, and optimizing battery configurations to elevate both efficiency and thermal stability [3]. Jin et al. developed a novel liquid cold plate for thermal management of EV batteries, demonstrating that an inclined structure could significantly enhance the cooling performance of the liquid cold plate [7]. Rao et al. proposed a thermal management system for cylindrical LIB blocks based on liquid cooling with a variable contact surface [8]. Their findings indicated that systems with a variable contact surface performed superiorly compared to those with a constant contact surface. Wang et al. investigated a BTMS using paraffin/aluminum foam composite phase change material for LIBs [9]. The results suggested that this material was effective in controlling the temperature rise during the discharge process of LIBs. Virendra et al. proposed a novel design of graphene-enhanced high thermal conductivity Composite Phase Change Materials (CPCMs) capable of delaying the onset of thermal runaway in LiNiMn-CoO₂ (NMC) and LiFePO₄ (LFP) cathode chemistries, thereby enhancing the safety of batteries [10]. Bais and colleagues developed a passive BTMS utilizing novel nano-enhanced phase change materials (NePCM) [11]. This system effectively reduces the operating temperature of LIBs under high-rate discharge conditions, thereby mitigating the risk of thermal runaway.

However, as the power and energy density of LIBs continue to increase, it is essential to implement effective temperature prediction and thermal diagnostics. This would enable timely adjustments to be made to charging rates or cooling strategies, preventing battery temperatures from reaching damaging levels. To this end, a deep understanding of battery thermal behaviors is imperative, along with the development of accurate thermal models for numerical simulations. During the charge/discharge process, heat generated internally within a battery needs to be transferred to the surface and subsequently dissipated into the surrounding air or through specific cooling methods. Residual energy stored within the battery could lead to cumulative temperature increases [12]. Thus, modeling the battery's thermal behavior involves addressing both the generation and transfer of heat, establishing corresponding energy equations and boundary conditions [13,14].

Nevertheless, considering that the heat generation process is closely linked to the rate of electrochemical reactions occurring inside the battery during charging and discharging, reliance solely on a thermal model is often inadequate for accurately describing and predicting the temperature rise characteristics during these processes [15]. Precise thermal modeling must always be integrated with physical modeling of the battery itself [16]. Therefore, BTMS need to collaborate closely with Battery Management Systems (BMS) to ensure the optimal performance

and safety of the battery. Moreover, considering that the BMS in EVs usually operate with limited computational resources, it is necessary to develop lightweight online battery electro-thermal coupled modeling techniques, as illustrated in Fig. 1. Such real-time and accurate implementations of thermal early warnings offers a novel thermal management approach, enabling further optimize battery charging and discharging strategies, driving plans, and cooling system control strategies through predictive battery operations and maintenance. Specifically, for the battery packs of vehicles, surrogate models can be used to simulate various driving plans in advance, allowing for the prediction of battery voltage and temperature under uncertain operating currents. This capability is crucial for enhancing the efficiency of online battery operations, as well as ensuring battery performance, safety, and lifespan.

LIBs are complex electrochemical systems where charging and discharging processes entail various reaction steps, including ion diffusion in electrolyte, lithium-ion intercalation in bulk electrode, and charge transfer between various interfaces, making it challenging to precisely describe their dynamic behaviors. These processes and thermal behavior of the batteries are interrelated closely with each other. Thus, a battery model which can precisely describe the above processes is prerequisite for battery thermal management. As illustrated in Fig. 2, in 1993 Newman et al. introduced the seminal Doyle–Fuller–Newman (DFN) model for porous electrodes, which provides a sophisticated and comprehensive mathematical description of the electrochemical process in lithium-ion batteries [17]. This model has been widely applied in simulations due to its accuracy across different battery types but entails numerous differential equations and substantial computational resources, hindering its deployment in practical BMS [18]. To reduce computational complexity, many subsequent studies aimed to simplify the model while preserving key physical mechanisms. For instance, the Single Particle Model (SPM) simplifies particle reactions in the electrode to a single representative particle and omits ion diffusion in the solution [19,20]. Such simplifications significantly improve computation speed, yet limitations in accuracy persist at higher current rates [21]. Moreover, the nonlinear relationship between state variables and terminal voltage hinders the online observability of the state variables, necessitating reliance on offline parameter identification [22]. On the other hand, the Equivalent Circuit Model (ECM), by aggregating complex electrochemical behavior into lumped elements, offers lower computational complexity and integrates advanced filtering techniques for online parameter identification and modeling, making it the most extensively used model in contemporary BMS systems [23]. However, traditional ECM approaches overlook nonlinear phenomena such as solid-phase diffusion in particles and electrochemical polarization, compromising high-precision modeling [24,25]. Therefore, balancing precision with computational load in physics-based models represents a notable challenge. Even with the most accurate models to date, a gap persists between the modeling and the actual systems.

In recent years, the rapid advancement and application of deep learning technologies have led to their integration into temperature monitoring and warning systems for EV batteries, aiming to enhance the accuracy and timeliness of predictions [17,26]. Table 1 summarizes recent research related to battery heat generation and temperature prediction. Zhu and colleagues developed a forecasting technique based on the Long Short-Term Memory (LSTM) algorithm that predicts temperatures at specific time points and accounts for long-term fluctuations in battery temperatures [27]. With a training-to-validation data ratio of 8:2, this model can achieve a remarkably low Root Mean Squared Error (RMSE) of 0.044 °C. Meanwhile, Hasan and his team introduced a data-driven model utilizing a Nonlinear Autoregressive with Exogenous inputs (NARX) neural network to estimate temperatures in lithium polymer batteries [28]. Findings suggest that NARX surpasses other complex methods such as wavelet, nonparametric regression, and fuzzy models in learning long-term dependencies by effectively mapping real data onto high-dimensional nonlinear spaces. In more than 95 % of the time, the RMSE of the estimated values is below 1 °C. Zafar et al. have designed an

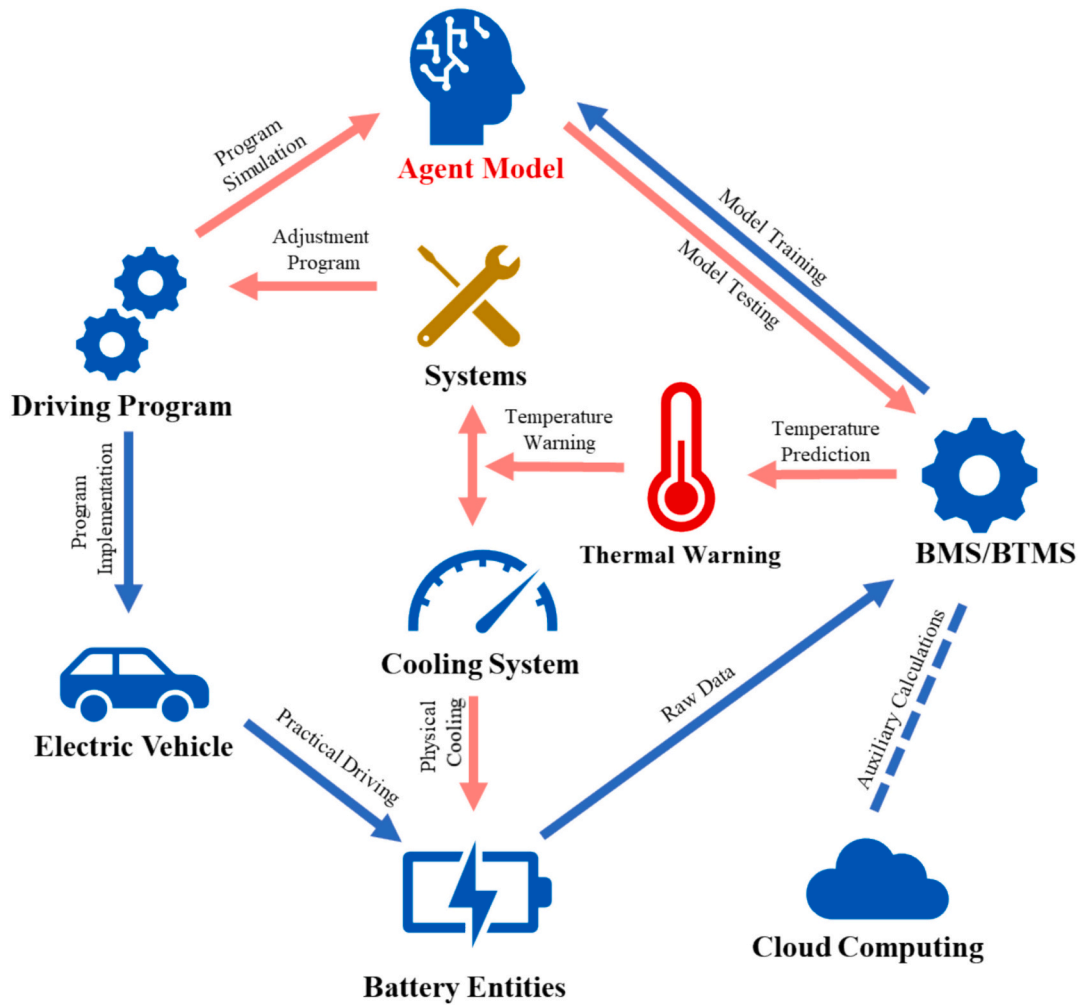


Fig. 1. Design of BTMS architecture for EVs based on online battery electric-thermal coupling modeling technology.

innovative deep learning architecture called DeepTimeNet, which integrates advanced deep learning techniques such as Convolutional Neural Networks (CNN), ResNet blocks, and Gated Recurrent Units (GRU) [29]. This architecture is adept at accurately modeling the time-dependent behavior of batteries, enabling precise estimation of lithium-ion battery surface temperatures under varying environmental conditions. Qi et al. have proposed a battery temperature prediction method featuring frequency domain reconstruction [30]. This approach combines Empirical Mode Decomposition, Long Short-Term Memory, and Attention Mechanism to effectively manage high-frequency non-stationary noise and mitigate prediction bias caused by changes in the battery's physical characteristics. These methods effectively incorporate a larger number of free parameters by mapping real data into a high-dimensional nonlinear space, facilitating easy and efficient parallel computation, all while reducing reliance on complex domain knowledge, thus promising to overcome the accuracy bottlenecks faced by traditional theoretical modeling. However, these techniques also require substantial datasets for training and face challenges in generalizing across different operational conditions due to a lack of domain-specific knowledge. Additionally, the often-opaque interpretability of these models can lead to user skepticism and decreased practical utility [31,32].

To leverage the strengths of both data-driven methods and physics-based models, recent academic research has shown a keen interest in electro-thermal hybrid models that integrate data-driven and physical approaches [26,41]. Feng et al. used simplified SPM coupled with aggregated thermal models to inform physical sub-models and then

employed feedforward neural networks to enhance accuracy under high currents and extreme temperatures, achieving a temperature estimation error of less than 0.7 °C within the range of 20–40 °C [33]. Tu et al. investigated various ways of combining SPM and Nonlinear Double-Capacitor (NDC) model with feedforward neural networks, achieving accurate voltage predictions with an RMSE of less than 10 mV [42]. Pang et al. developed a thermodynamics-inclusive Single Particle Model with Thermal (SPMT) effects to characterize LIBs' thermal behavior, inputting solid surface concentrations as physical information into a Bidirectional-LSTM network for thermal estimation in LIBs, with a Mean Absolute Error (MAE) of only 0.542 kW/m³ under Dynamic Stress Test (DST) conditions [34]. Liu et al. have developed an innovative hybrid framework that integrates a battery equivalent circuit model with a transformer-based neural network [35]. This integration facilitates the offline training of hyperparameters, thereby enhancing the accuracy of real-time temperature predictions. These methods mainly use state information from physical models and physics-based predictions to bridge the “physics gap” in neural networks; however, since they are grounded in static physical foundations and often rely on offline parameter identification, they may not always be practicable in real-time operations. Consequently, the effectiveness and fidelity of these models can diminish over time with LIB aging. Despite their vast potential, the intersection of deep learning, physical modeling, and battery thermal management is still emerging and calls for comprehensive research in developing reliable models, collecting integrative data, and incorporating deep learning into existing systems.

Physics-Informed Neural Networks (PINN) represent a novel

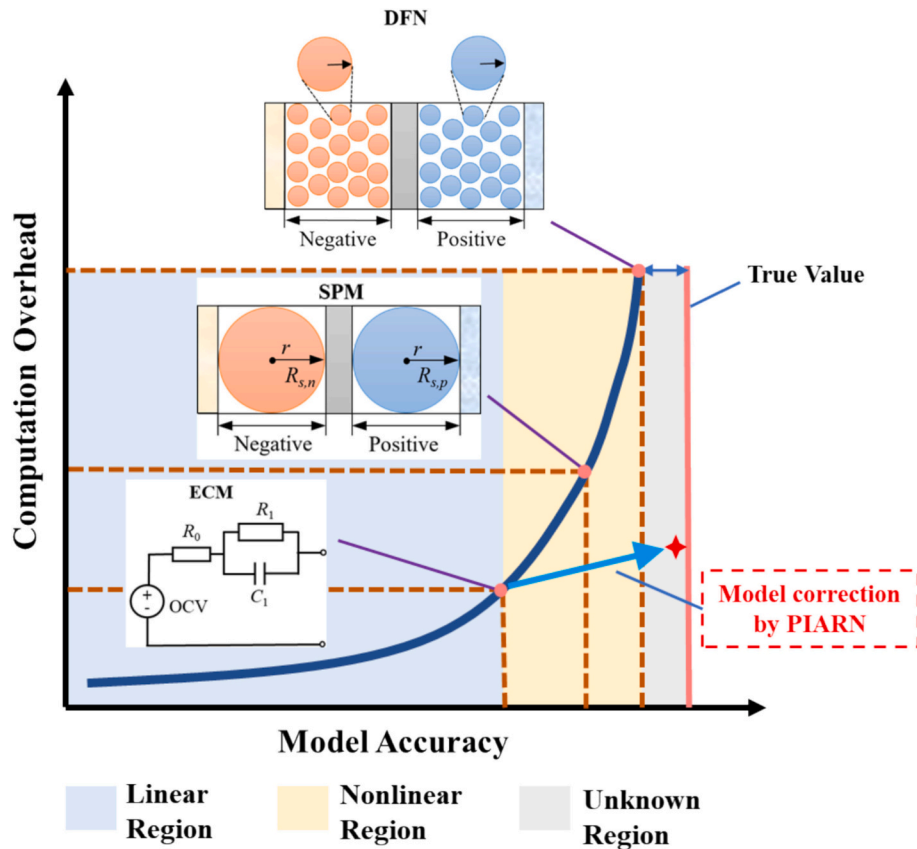


Fig. 2. Comparison of different physics-based LIB models and the research objectives of this study.

Table 1

A comprehensive review of recent literature on LIB heat generation and temperature prediction.

References	Methods	Category	Operating condition	Results
Ref [27]	LSTM	data-driven	long cycle	0.044 °C RMSE
Ref [28]	NARX	data-driven	photovoltaic energy storage	less than 1 °C RMSE
Ref [29]	DeepTimeNet	data-driven	random discharging	0.097 °C RMSE
Ref [30]	EMD-LSTM-AM	data-driven	random discharging	0.31 °C maximum error
Ref [33]	ETNN	hybrid model	random discharging	0.68 °C RMSE at 20 °C
Ref [34]	PINN	hybrid model	random discharging	0.542 kW/m ³ MAE
Ref [35]	Transformer-based	hybrid model	random discharging	0.044 °C RMSE
Ref [36]	Elman NN	data-driven	constant current discharging	0.6969 °C MAE
Ref [37]	GRU-RNN	data-driven	random discharging	0.75 °C maximum error
Ref [38]	Random Forest	data-driven	random discharging	0.9023 °C RMSE
Ref [39]	DT LSTM	data-driven	constant current discharging	0.17 °C RMSE
Ref [40]	BINN	hybrid model	constant current discharging	1.076 °C MSE

paradigm that integrates physical systems with neural networks, enabling the explicit encoding of physics-based domain knowledge into machine learning. This facilitates dynamic updating of hybrid models

using limited online observational data. Currently, PINNs are primarily applied in battery modeling and voltage prediction. Felipe A.C. and colleagues investigated the viability of physics-guided Recurrent Neural Networks (RNNs) along with the optimization of underlying physical parameters through backpropagation algorithms while incorporating data-driven nodes to address uncertainties in the physical formulation [43]. Renato G., Wang, and others embedded reduced-order electrochemical models [44,45] and second-order ECMs [46] of LIBs into physics-guided RNNs, establishing physics-driven implicit constraints, with multilayer perceptron characterizing the nonlinear associations between internal states and observational quantities. However, introducing deep neural network nodes into the RNN architecture significantly increases the sequence length and model depth, leading to prolonged training times and slower convergence due to gradient vanishing in long-sequence prediction problems, thereby hindering effective accuracy improvement. Consequently, existing physics-data hybrid models, either reliant on parameter-based offline identification or constrained by training efficiency, struggle to be directly applied in real-world dynamic scenarios that demand rapid response and online updates.

To address the aforementioned issues, it is necessary to design a more efficient network architecture that leverages the parallel computing capabilities and nonlinear mapping abilities of data-driven methods while retaining the main physical mechanisms to balance computational complexity and accuracy (as illustrated in Fig. 2). Here, we propose a new network architecture named Physics-Informed Attention Residual Network (PIARN) which incorporates a residual neural network to correct the LIB model, offering significant improvements in thermal warning performance for EV batteries. Compared to traditional deep learning approaches, PIARNs, by integrating physical prior knowledge, can more accurately simulate and predict battery thermal behavior under various operating conditions. The primary contributions of this

work are as follows:

1. Constructed an electro-thermal hybrid reduced-order model based on the NDC model and further embedded it into the physics-guided RNN. This model design preserves the primary physical mechanisms while reducing the serial computational burden of RNN architectures, facilitating the updating and learning of physical parameters through backpropagation algorithms to enhance interpretability and generalization capabilities. Additionally, a novel Adaptive Weighting (AW) adjustment method was designed to address the magnitude differences between loss functions, improving convergence efficiency.
2. Compensated for and corrected the limitations in the accuracy of physics modeling through the architecture design of the PIARN. The residual attention network block first concatenates the input-output and state information of the physics differential network, utilizes physics prior knowledge for physics normalization, then fuses information between different features through the Efficient Channel Attention (ECA) block, and finally learns the dynamic variation pattern of residuals using the Bidirectional Gated Recurrent Unit (Bi-GRU) temporal block.
3. Demonstrated the efficiency and accuracy of PIARN modeling on multiple dynamic datasets including experimental, simulation, and publicly available National Aeronautics and Space Administration (NASA) datasets. Through various comparative and ablation experiments, the effectiveness of the method design was validated. Ultimately, a potential online iterative training strategy based on this method is proposed, illustrating its precision and robustness in temperature prediction and thermal warning throughout the entire lifecycle of LIBs and under varying environmental conditions.

The remaining sections of this report are organized as follows: **Section 2** introduces the PIARN framework proposed in this paper, including the physics block and residual block. **Section 3** presents the three dynamic discharging datasets used in this study. **Section 4** conducts case studies and comparisons of the PIARN method and further designs an online iterative training framework. Finally, major conclusions and insights are discussed in **Section 5**.

2. Methods

The proposed PIARN method in this study comprises blocks such as the physical differential block and the residual attention block. This section firstly presents the theoretical foundations and functions of each block, followed by an overview of the overall network architecture of PIARN and the design principles behind its loss function.

2.1. Physical differential block design

To integrate physical knowledge into the PIARN architecture, this paper introduces a novel physical differential block. Within this block, a physics-guided RNN is implemented, which represents a specialized network within the framework of PINNs that embeds physical knowledge in the form of differential equations directly into its core. This design facilitates the incorporation of prior knowledge and the propagation of physically-based temporal information, thereby, it has been designated as the Physics-Informed Recurrent Neural Network (PIRNN) in this study. **Fig. S1** delineates the interrelations of these key concepts. By leveraging partial known physical information, PIRNN enhances the temporal correlations within the network, capturing time dependencies and dynamic variations between limited observational data. It finds applications in fields such as time series analysis, signal processing, and control systems [44,47,48]. The design of this model is particularly advantageous for integration with traditional neural network models, allowing gradient descent and parameter optimization through backpropagation algorithms. Consequently, we have employed the PIRNN to

construct a physical differential block for a reduced-order model of LIB electro-thermal coupling.

The research on PIRNN aims to leverage the characteristics of information propagation in physical systems and apply them to neural network models, thereby enhancing the modeling capabilities and predictive performance for time series data and dynamic systems.

2.1.1. Electro-thermal coupling model

To achieve optimal generalization performance at lower computational complexity, the ideal reduced-order battery model should include linear principal components of the mechanisms in each component. Traditional ECMs often employ multiple RC circuits to characterize the capacitance effects and ion diffusion effects with different time constants. Compared to electrochemical models, although greatly simplified in structure, they lack representation of solid-phase diffusion processes and their dependency on Open Circuit Voltage (OCV). Recently, Tian et al. proposed a novel equivalent circuit model NDC, which maps the solid-phase diffusion processes of the positive and negative electrodes into a coupled dynamic circuit, thus achieving higher fidelity [49]. Therefore, this study is based on the NDC model for physical modeling and incorporates adaptive modifications within the overall framework while coupling with a thermal model.

The modified NDC electro-thermal coupling model, as depicted in **Fig. 3**, primarily consists of an equivalent electrical circuit model and a thermal model. Within the thermal model, the heat source Q is determined by the operating state of the equivalent electrical circuit model. The equivalent circuit model, a modified NDC model, consists of two coupled parts. The first part simulates the solid-phase ion diffusion process in the electrode, which includes two parallel RC circuits. R_b - C_b represents the inner region of the electrode particles, while R_s - C_s represents the surface region of the electrode. Here, $R_s \ll R_b$, and typically R_s can be set to 0, with R_b reflecting the diffusion rate coefficient of lithium ions between the inner and outer layers. C_b and C_s reflect the total amount of lithium ions that can be accommodated in the inner and surface regions, respectively. Their relative sizes depend on the volume ratio, β , between the inner and outer layers (similar to the specific surface area of particles). Correspondingly, V_b and V_s represent the normalized virtual potentials of the inner and outer layers, respectively. Assuming that the change in particle volume ratio during battery aging can be neglected, C_b and C_s can be expressed in terms of the charge capacity $q_{\max}$, i.e.,

$$\beta = \frac{C_b}{C_s} \quad (1)$$

$$C_b = \frac{\beta}{\beta + 1} q_{\max} \quad (2)$$

$$C_s = \frac{1}{\beta + 1} q_{\max} \quad (3)$$

Assuming that the actual charge stored in the inner and outer layers of the particle are q_b and q_s , respectively, the virtual potential can be represented by the charge concentration as follows:

$$V_b = \frac{q_b}{C_b} \quad (4)$$

$$V_s = \frac{q_s}{C_s} \quad (5)$$

The second part simulates the dynamic voltage output of the battery, including a voltage source U , resistance R_0 , and a series R-C pair. The resistance R_0 represents the combination of contact resistance, solution resistance, and electrochemical polarization resistance, among other ohmic resistances. It is worth noting that the electrochemical polarization resistance, governed by the Butler-Volmer equation, is a nonlinear resistance related to the interface current density and battery State of

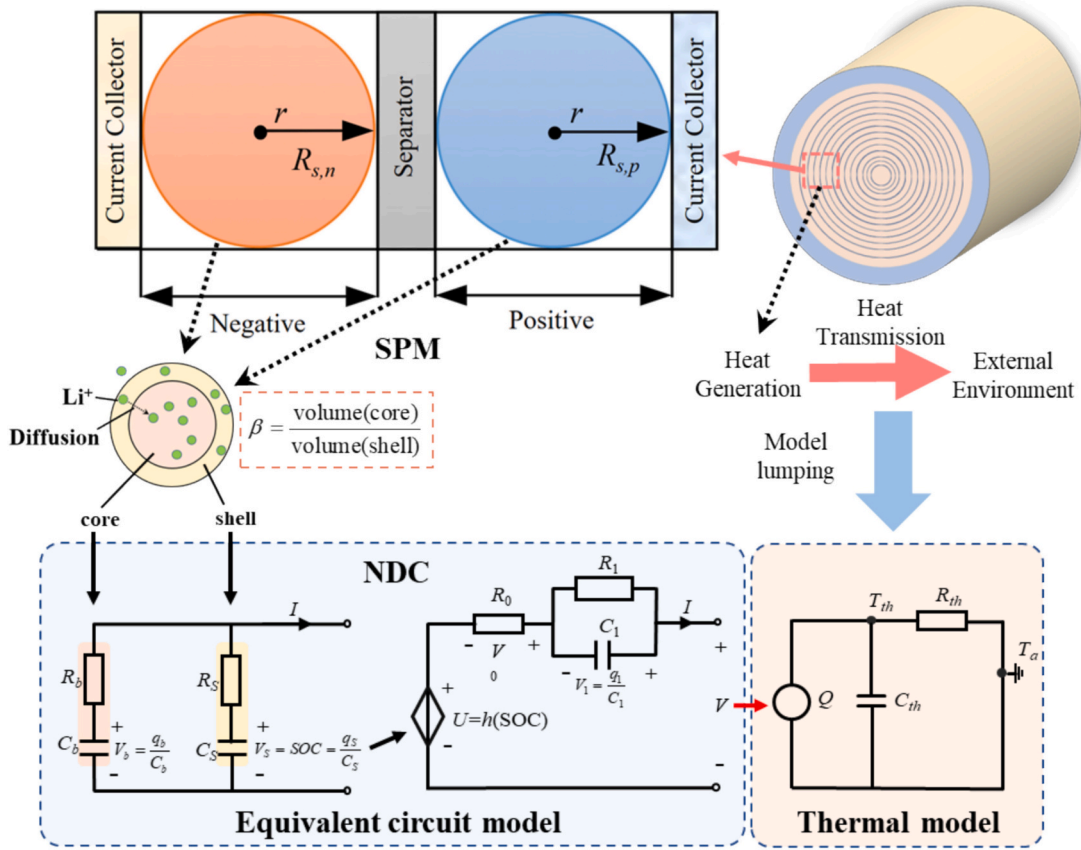


Fig. 3. Modified NDC electro-thermal coupling model.

Charge (SOC) [22,50], here we consider only its linear principal component. R_1 - C_1 models the liquid-phase diffusion process of the particles. The controlled power source $U = h(\text{SOC})$ represents the open-circuit potential. It is noteworthy that the SOC is traditionally defined as the ratio of the total charge within the electrode to its overall capacity. However, strictly speaking, the open circuit potential has a stronger correlation with the surface charge concentration of the electrode particles. Therefore, we represent the SOC using the virtual potential, V_s .

Correspondingly, for temperature estimation, we adopt a first-order lumped thermal model for its computational efficiency [51,52]. As shown below:

$$C_{th} \frac{dT_{th}}{dt} = \frac{T_a - T_{th}}{R_{th}} + I(U - V) + I \left(T \frac{\partial U}{\partial T}(\text{SOC}) \right) \quad (6)$$

In the equation, C_{th} represents heat capacity, R_{th} represents thermal resistance, T_{th} represents the battery lumped temperature, t represents time, I represents current, U represents open circuit potential, V represents terminal voltage, T_a represents ambient temperature, and $\frac{\partial U}{\partial T}(\cdot)$ represents the entropy function of the battery reaction. On the right side of the equation, the first term represents the heat dissipation rate, while the second and third terms represent the irreversible ohmic heating and reversible reaction heat generation respectively, which correspond to the heat source Q in Fig. 3. $\frac{\partial U}{\partial T}(\cdot)$ is a nonlinear function related to SOC and temperature, typically requiring offline identification. Therefore, in physical modeling, we only consider linear ohmic heat.

The equations of state for the modified NDC electro-thermal coupling model are as follows [49,51]:

$$\begin{aligned} \frac{dq_s}{dt} &= \left(-\frac{q_s}{C_s} + \frac{q_b}{C_b} \right) / R_b \\ \frac{dq_b}{dt} &= \left(\frac{q_s}{C_s} - \frac{q_b}{C_b} \right) / R_b + I \\ \frac{dq_1}{dt} &= -\frac{q_1}{R_1 C_1} + I \\ \frac{dT_{th}}{dt} &= \frac{T_a - T_{th}}{R_{th} C_{th}} + I \left(IR_0 + \frac{q_1}{C_1} \right) / C_{th} \end{aligned} \quad (7)$$

The corresponding observation equations are as follows:

$$\begin{cases} V = h\left(\frac{q_s}{C_s}\right) - \frac{q_1}{C_1} - IR_0 \\ T = T_{th} \end{cases} \quad (8)$$

In the equation, C_s and C_b are expressed by $q_{\max}$ and β through Eqs. (2) and (3). Here, β serves as a fixed hyperparameter, which will be further optimized using grid search. As a result, the simplified electrochemical-thermal coupling model only contains a small number of unknown parameters $\theta = [q_{\max}, R_0, R_b, R_1, C_1, R_{th}, C_{th}]$, and it is expected to capture the main trends with lower computational complexity.

2.1.2. PIRNN structure

Real-world data is typically obtained as discrete sampled values at regular time intervals, while RNNs have demonstrated powerful capabilities in handling the transmission of dynamic time series information [47,53]. Therefore, we transform the continuous-time system mentioned earlier into a differenced discrete-time system to better embed the physical essence into the architecture of RNNs, as shown below:

$$\begin{aligned}
q_s(t) &= \left(-\frac{q_s(t-1)}{C_s} + \frac{q_b(t-1)}{C_b} \right) \frac{\Delta t}{R_b} + q_s(t-1) \\
q_b(t) &= \left(\frac{q_s(t-1)}{C_s} - \frac{q_b(t-1)}{C_b} \right) \frac{\Delta t}{R_b} + I(t) + q_b(t-1) \\
q_1(t) &= \left(-\frac{q_1(t-1)}{R_1 C_1} + I(t) \right) \Delta t + q_1(t-1) \\
T_{th}(t) &= \frac{T_a - T_{th}(t-1)}{R_{th} C_{th}} \Delta t + I(t) \left(I(t) R_0 + \frac{q_1(t-1)}{C_1} \right) \frac{\Delta t}{C_{th}} + T_{th}(t-1)
\end{aligned} \quad (9)$$

$$\begin{cases} V(t) = h\left(\frac{q_s(t)}{C_s}\right) - \frac{q_1(t)}{C_1} - I(t)R_0 \\ T(t) = T_{th}(t) \end{cases} \quad (10)$$

Embedding the difference equation into a RNN architecture yields the PIRNN physics differential block illustrated in Fig. 4, where the hidden state variable denoted as $H(t) = [q_s(t), q_b(t), q_1(t), T_{th}(t)]$ is sequentially propagated through the recurrent nodes to capture the dynamic characteristics of the physical system. Given that the non-linear OCV-SOC curves are relatively stable and mainly dependent on the materials themselves, with minimal influence from individual battery variations and aging levels, as shown in Figs. S2 and S3. Therefore, the OCV-SOC relationship curve can be shared among similar batteries, typically requiring only a single offline measurement and fitting through a polynomial function of SOC. To obtain more precise data, a simple neural network with two hidden layers, each comprising 64 neurons is employed to fit the OCV-SOC relationship for similar batteries (as depicted in Fig. 5). And the parameters of this network are frozen before embedding it into the PIRNN. As a result, this network does not participate in subsequent gradient descent, which benefits the stability of the physical architecture and preservation of prior knowledge.

It is worth noting that although learning physical parameters can be

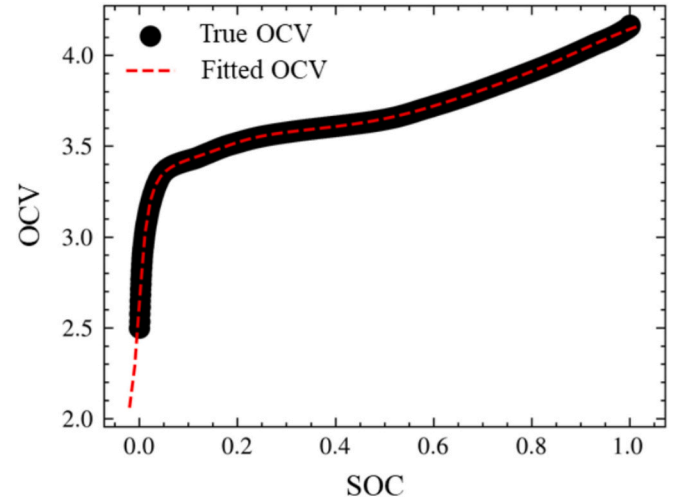


Fig. 5. Comparison of true OCV and fitted OCV.

achieved using the backpropagation algorithm within this framework, the presence of some non-linear mappings in the state space does not guarantee its global optimization capability. Therefore, setting reasonable initial values for the physical parameters remains crucial. In order to enhance the reflection of physical parameters on the actual dynamics of the battery, as well as improve the convergence and prediction accuracy of the model, we incorporate extensive experimental observations and literature [20,22] to define the relevant initialization parameters, as shown in Table S1. Among them, the initial value of q_{max} is set to the rated capacity of the battery. The values for R_0, R_b, R_1, C are determined based on a combination of reference literature and empirical experiments [49]. The values for R_{th} and C_{th} are obtained by fitting the

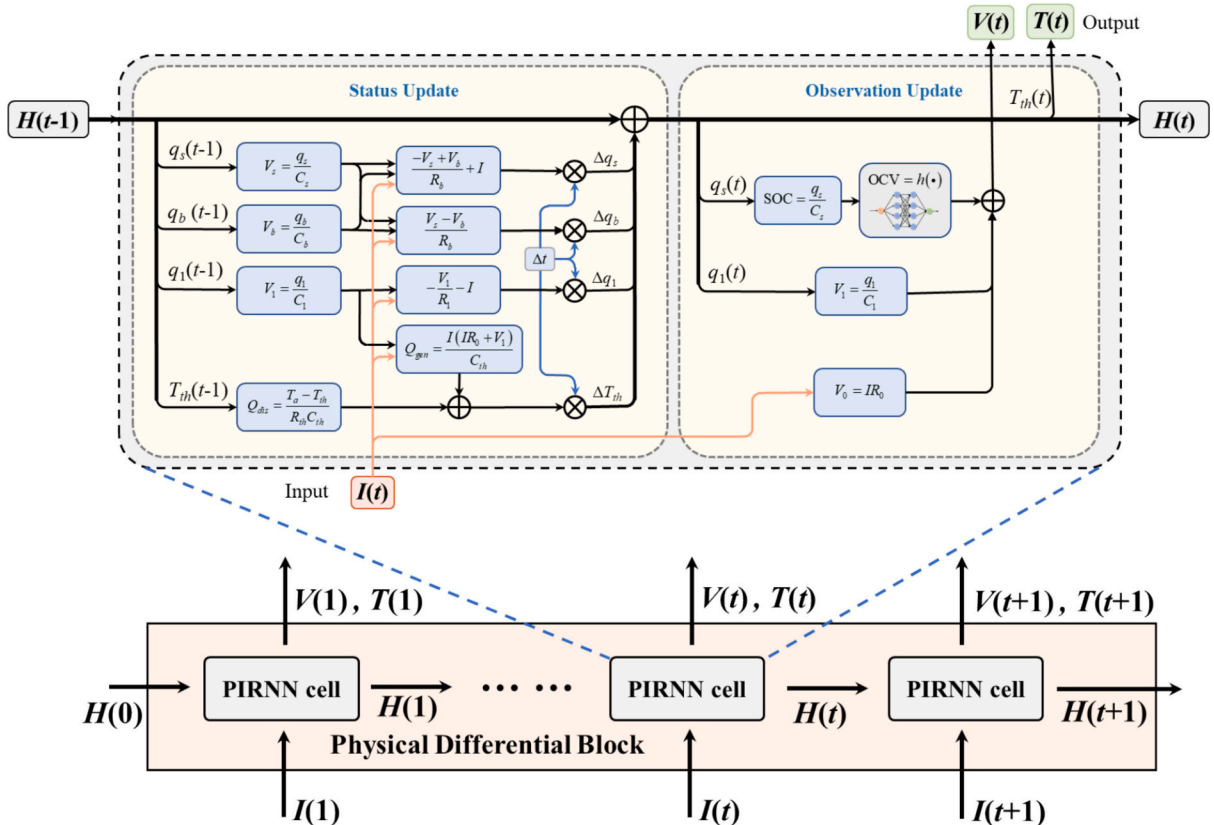


Fig. 4. Physics differential block composed of PIRNN based on NDC electro-thermal coupling model.

curve of temperature decrease during the resting period in the battery temperature profile from the experiment dataset (as introduced in **Subsection 3.1**). Therefore, the model proposed in this paper does not rely on precise offline parameter identification. Except for the parameters $q_{\max}$ and the SOC-OCV curve relationship, all other parameters can be shared across different battery systems. This characteristic enhances the model's applicability and generalizability.

2.2. Residual attention block design

Deep learning has demonstrated powerful non-linear mapping capabilities in multi-dimensional time series problems and has been widely applied in fields such as image recognition and natural language processing [54]. Therefore, we utilize deep neural networks to bridge the physical gap between the reduced-order physics differential block and the actual dynamics. In this study, we concatenate the input, output, and hidden state of the physical differential module to form the input to the residual attention module, denoted as $X^{\text{Res}} = [V, T, q_s, q_b, q_1, I]$.

2.2.1. Physical normalization layer

To enhance the training effectiveness and generalization capability of the neural network model, it is common practice to apply layer normalization or batch normalization before the hidden layers of the neural network for input data standardization. Here, considering that the output of the physics differential block holds specific physical meaning and there may be variations among different samples, normalizing the input data could potentially compromise the richness of information. Therefore, leveraging prior knowledge, we establish shared upper and lower limits for different samples to ensure neural network stability and efficiency during training. The upper and lower limits for various states and observation quantities are as follows:

$$I \in [0, \alpha q_{\max}] \quad (11)$$

$$V \in [V_{lb}, V_{ub}] \quad (12)$$

$$T_{th} \in [T_a, T_a + 10] \quad (13)$$

$$q_s \in \left[0, \frac{1}{\beta + 1} q_{\max}\right] \quad (14)$$

$$q_b \in \left[0, \frac{\beta}{\beta + 1} q_{\max}\right] \quad (15)$$

$$q_1 \in [0, C_1] \quad (16)$$

where V_{lb} and V_{ub} represent the lower and upper voltage bounds of the battery, respectively, and T_a denotes the room temperature. The upper bound is empirically set as $(T_a + 10)$. Furthermore, the upper bound of I is determined by a multiple α of the current capacity $q_{\max}$. In this study, based on the dataset, we set α to 2.5.

The normalization formula is

$$x = 2 \frac{x - x_{lb}}{x_{ub} - x_{lb}} - 1 \quad (17)$$

It is worth noting that the physical normalization layer here does not limit the actual bounds of certain variables, such as temperature, which can exceed the upper bound in some scenarios.

2.2.2. Channel attention block

In recent years, the channel attention mechanism has shown significant potential in boosting the performance of Deep Convolutional Neural Networks (DCNN) and is increasingly being applied to fuse temporal feature information. In comparison to the classical Squeeze-and-Excitation (SE) channel attention block [55], the ECA block effectively avoids the dimensionality reduction operation of fully connected layers, preserving the correspondence between channels and weights

[56]. While maintaining computational efficiency, it enables the network to adaptively learn the importance of each channel, thus enhancing feature representation capability. Therefore, this study utilizes the ECA block to capture the correlations between different sequential features.

As illustrated in Fig. 6, the initial input dimension is $C_0 \times S \times 1$, where C_0 represents the number of features in X^{Res} , and S represents the length of the time series. To enhance the data representation capability, the original input is expanded through an embedding layer, resulting in a tensor of dimensions $C \times S \times 1$. Subsequently, it is compressed via a global average pooling to $C \times 1 \times 1$, where C denotes the feature dimension of the ECA hidden layer, henceforth denoted as D_{att} in the subsequent text. Subsequently, the size of the adaptive convolutional kernel, $k = \left\lceil \frac{\log_2(C)+1}{2} \right\rceil$, is calculated, where C represents the number of input channels. One-dimensional convolution is used to compute channel weights, followed by applying a Sigmoid activation function $\sigma(\cdot)$ to map the weights between 0 and 1. The transposed weight values are multiplied by the previously expanded feature matrix of dimensions $C \times S \times 1$, resulting in feature maps under varying weights.

2.2.3. Time series block

In addition to fusing information between residual features, we also need to capture the dynamic temporal dependencies of multi-dimensional residual information over time. To achieve this, we utilize a bidirectional information propagation function, Bi-GRU, which is an enhanced RNN structure [57]. By incorporating update gates and reset gates to combine input parameters, hidden states, and previous states in controlling output information, Bi-GRU can efficiently capture dependencies in long temporal sequences.

The computational formula for the basic GRU (Gated Recurrent Unit) cell is as follows [58]:

$$z_t = \sigma(W_z \times |h_{t-1}, x_t| + b_z) \quad (18)$$

$$r_t = \sigma(W_r \times |h_{t-1}, x_t| + b_r) \quad (19)$$

$$\hat{h}_t = \tanh(W_h \times (r_t \times h_{t-1}, x_t) + b_h) \quad (20)$$

$$h_t = (1 - z_t) \times h_{t-1} + z_t \times \hat{h}_t \quad (21)$$

where z_t and r_t respectively denote the outputs of the update gate and the reset gate; x_t represents the input at the current timestep; h_{t-1} is the hidden state from the previous timestep; W_z , W_r , and W_h are the weight matrices for the update gate, reset gate, and new candidate hidden state, respectively; b_z , b_r , and b_h are the corresponding biases; $\sigma(\cdot)$ refers to the sigmoid activation function; $\hat{h}_t$ is the new candidate hidden state; and h_t is the hidden state at the current timestep.

The Bi-GRU model captures the bidirectional dependencies of sequence data by combining two independent GRU units—one forward and one backward—thereby enhancing the model's ability to process information from both past and future contexts, i.e.

$$H_t^c = \left[\vec{h}_t, \overleftarrow{h}_t \right] \quad (22)$$

In this context, $\vec{h}_t$ and $\overleftarrow{h}_t$ respectively represent the hidden states of the forward and backward GRU units, while H_t^c denotes the hidden state vector of the Bi-GRU.

Following the Bi-GRU, a shared fully connected layer is used for dimensionality reduction. The fully connected architecture comprises one input layer, two hidden layers, and one output layer. The computations for the hidden and output layers are performed using the following formulas [59]:

$$Y_m = \text{ReLU}(W_m D_m + b_m) \quad (23)$$

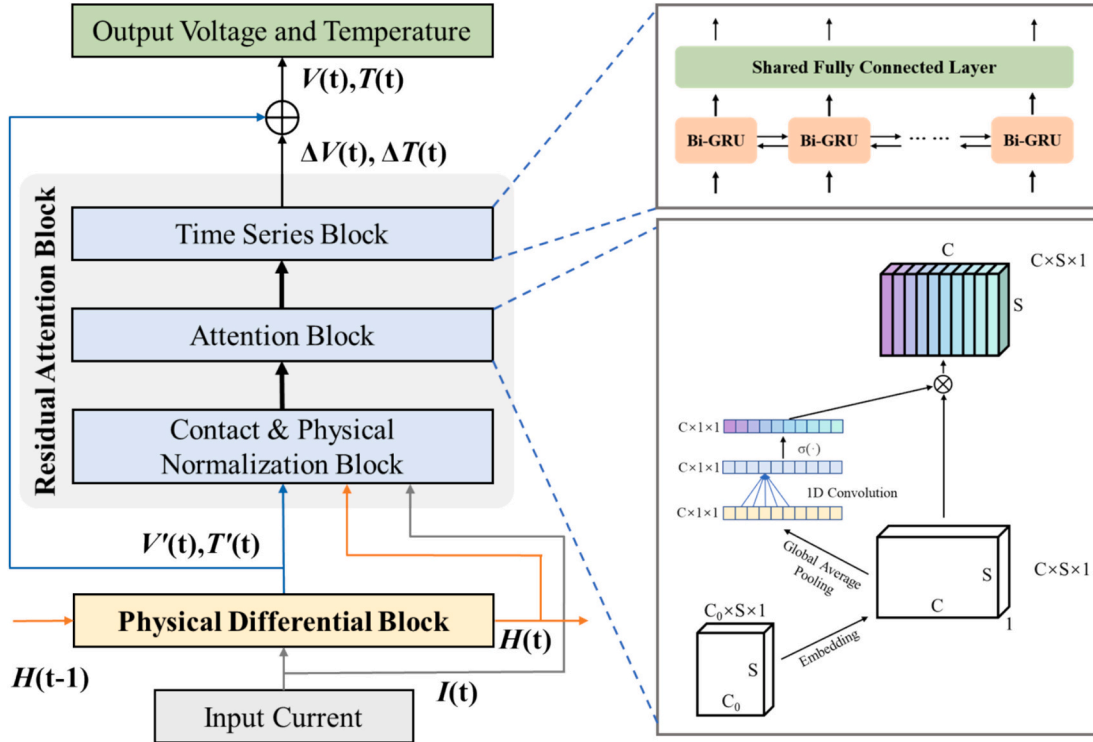


Fig. 6. Physical information attention residual network architecture.

In the formula, m represents the number of layers, while D_m , W_m , b_m , and Y_m correspond to the input, weights, biases, and output of the m -th layer, respectively. In the final layer, Y_m is reduced to two dimensions, corresponding to ΔV_t and ΔT_t . In this study, the dimension of the hidden layers in the shared fully connected layer is set to 32.

2.3. Architectural design

2.3.1. Overall architecture design program

Building upon the primary physical mechanisms of the PIRNN and non-linear adjustments from the residual attention network, we have developed a novel PIARN to create an integrated model for LIBs, as illustrated in Fig. 6. The input is a time series of electrical current, processed by the physics differential block to derive state variables and physical outputs of voltage and temperature at each time step, as shown in Eq. (24). These parameters—current, voltage, temperature, and state variables—are concatenated and fed into a physical normalization block. This data is then mapped to a higher-dimensional space via an encoder, facilitating deeper semantic acquisition through representation learning. Subsequent attention and temporal blocks synergize channel and temporal features, with the high-dimensional sequence being projected onto a two-dimensional plane using fully connected layers to capture residual representations of voltage and temperature. Lastly, global output is produced by summing the physical output with the residuals, as shown in Eq. (25).

$$[V(t), T(t), H(t)] = \text{Phy}[I(t), H(t-1)] \quad (24)$$

$$[V(t), T(t)] = \text{Res}[V(t), T(t), H(t), I(t)] + [V(t), T(t)] \quad (25)$$

where $\text{Phy}(\cdot)$ and $\text{Res}(\cdot)$ represent the abstract functions of the physical differential module and the residual attention module, respectively; $V(t)$ and T_t denote the voltage and temperature outputs from the physical module.

While the concept of residual learning has been integrated into hybrid physical machine learning modeling, it typically relies on a

predefined physical foundation [33,42]. Indeed, residual networks based on residual learning have gained widespread usage in deep learning, originating from He Kaiming's ResNet [60]. These networks incorporate skip connections, allowing direct transfer of information across different layers, thus preserving more gradient information and profoundly influencing the development of deep learning. Actually, such a design is also beneficial for maintaining dynamic physical underpinnings, thereby facilitating the integration and online updating of the physical foundation with residual networks.

2.3.2. Design of the loss function

The loss function of the PIARN consists of multiple loss components, generally expressed as follows:

$$\ell = \ell_V + \ell_T + \ell_B \quad (26)$$

where ℓ_V , ℓ_T and ℓ_B indicate the loss terms for voltage observation error, temperature observation error, and boundary condition error, respectively. The specific expressions for these loss terms are as follows:

$$\ell_V = \frac{1}{S} \sum_{t=1}^S (V_{\text{pred}}(t) - V_{\text{true}}(t))^2 \quad (27)$$

$$\ell_T = \frac{1}{S} \sum_{t=1}^S (T_{\text{pred}}(t) - T_{\text{true}}(t))^2 \quad (28)$$

$$\ell_B = \frac{1}{S} \sum_{t=1}^S [\text{ReLU}(V_{\text{pred}}(t) - V_{ub}) + \text{ReLU}(V_{lb} - V_{\text{pred}}(t)) + \text{ReLU}(-q_s(t)) + \text{ReLU}(-q_b(t))] \quad (29)$$

where the boundary term related to $V_{\text{pred}}(t)$ aims to constrain the upper and lower limits of terminal voltage $V_{\text{pred}}(t) \in [V_{lb}, V_{ub}]$, while the boundary term associated with $q_s(t)$ and $q_b(t)$ is designed to ensure non-negative capacity, preventing system oscillations and non-convergence during training due to initialization parameter biases. A ReLU activa-

tion function is employed to impose penalty for exceeding boundaries, with this term being zero when not exceeded.

2.3.3. Adaptive loss weighting

It is worth noting that there is a considerable difference in magnitude between the output physical quantities of voltage and temperature, which may result in the loss function failing to guide the correct gradient descent direction, leading to inadequate training for certain tasks. In order to dynamically adjust and balance the relative sizes of these two quantities, avoiding the domination of training process by a single task, we aim to apply dynamically varying weights to these two loss components. Therefore, an AW technique is proposed, with the specific process outlined in Fig. 7 and the corresponding pseudocode presented in Algorithm 1.

The improved loss function is formulated as:

$$\ell = w_1 \ell_V + w_2 \ell_T + \ell_B \quad (30)$$

guiding a slow change in weights in the later stages. Thus, we update μ using the formula:

$$\mu^i = \xi_1 + (\xi_N - \xi_1) \left(1 - \exp\left(-\frac{i}{T_r}\right) \right) \quad (33)$$

where ξ_1 is the initial decay factor value (set to a larger value of 0.9 to ensure smooth weight changes in the initial stages) and ξ_N is the final decay factor value (set to 0.999 to indicate a slow change rate in weights in the later stages of iterations). T_r is the decay factor shaping temperature used to control the speed of decay. Empirically, we found that setting $T_r = 100$ leads to better convergence results. Additionally, when achieving good convergence, the boundary condition loss component ℓ_B approaches zero, and therefore, no weight needs to be assigned to this loss component.

Algorithm 1. AW method based on dynamic weight decay factors

Algorithm 1: AW method based on dynamic weight decay factors

Input: $I, V_{\text{true}}, T_{\text{true}}$

Output: The trained PIARN model

```

1: Initialize  $w_1 = w_2 = 0.5, i = 1$ 
2: while  $i \leq N_{\text{epoch}}$  do
3:    $V_{\text{pred}}, T_{\text{pred}} \leftarrow$  Forward operation
4:    $\ell_V, \ell_T \leftarrow$  Calculation of loss terms
5:    $\mu \leftarrow$  Updating the weight decay factor
6:   if  $w_1 \ell_V > w_2 \ell_T$  then
7:      $w_1 = w_1 \mu, w_2 = 1 - w_1$ 
8:   else
9:      $w_2 = w_2 \mu, w_1 = 1 - w_2$ 
10:  end if
11:   $\ell \leftarrow$  Calculation of the total weighted loss
12:   $i \leftarrow i + 1$ 
13:  Updated network parameters  $\leftarrow$  Loss backpropagation
14: end while
15: return the trained model

```

Here, w_1 and w_2 are adaptive weights for ℓ_V and ℓ_T respectively, both initialized to 0.5.

Specifically, in each iteration, a dynamic shrinkage factor μ is applied to the weight of the larger value between $w_1 \ell_V$ and $w_2 \ell_T$. For example, in the i -th iteration, if $w_1 \ell_V > w_2 \ell_T$, then

$$\begin{aligned} w_1^i &= w_1^{i-1} \mu^i \\ w_2^i &= 1 - w_1^i \end{aligned} \quad (31)$$

if $w_1 \ell_V > w_2 \ell_T$, (then)

$$\begin{aligned} w_2^i &= w_2^{i-1} \mu^i \\ w_1^i &= 1 - w_2^i \end{aligned} \quad (32)$$

Here, $\mu \in (0, 1]$ and its optimal value should facilitate a rapid balance of weights for loss components in the early stages of iterations, while

2.3.4. Evaluation metrics

In this study, the performance of the model is evaluated using the metrics of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), and coefficient of determination (R^2) [61,62].

$$\text{MAE} = \frac{1}{S} \sum_{t=1}^S |Y_{\text{pred}}(t) - Y_{\text{true}}(t)| \quad (34)$$

$$\text{RMSE} = \sqrt{\frac{1}{S} \sum_{t=1}^S (Y_{\text{pred}}(t) - Y_{\text{true}}(t))^2} \quad (35)$$

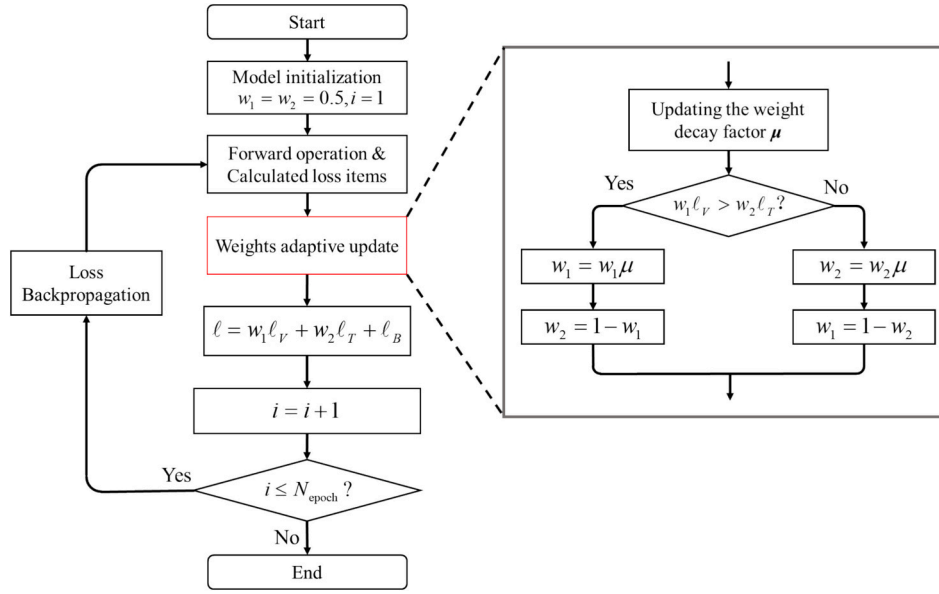


Fig. 7. Flowchart of AW method based on dynamic weight decay factor.

$$R^2 = 1 - \frac{\sum_{t=1}^S (Y_{\text{pred}}(t) - Y_{\text{true}}(t))^2}{\sum_{t=1}^S \left(Y_{\text{pred}}(t) - \frac{1}{S} \sum_{t=1}^S Y_{\text{pred}}(t) \right)^2} \quad (36)$$

where Y represents the output of V or T .

3. Dataset introduction

This paper focuses on the voltage and temperature variations of LIBs under random discharge conditions. To comprehensively evaluate our PIARN model, we utilized three random discharge datasets, consisting of experimental data (ExpData), simulation data (SimData), and the NASA

random dataset (NASADData). The following provides a detailed introduction to these datasets. In the subsequent sections, the four batteries of each group are sequentially labeled as Bat1 through Bat4. In this study, to accommodate the different battery types across datasets, we utilize the C-rate to measure the battery's charge/discharge rate, which is defined as the ratio of the charge/discharge current to the rated capacity. For instance, for a battery with a rated capacity of 5.0 Ah, 1C and 2C correspond to charge/discharge currents of 5 A and 10 A, respectively.

3.1. Dataset 1: Experimental data

To collect experimental data on voltage and battery temperature under various operating conditions, an experimental platform as shown

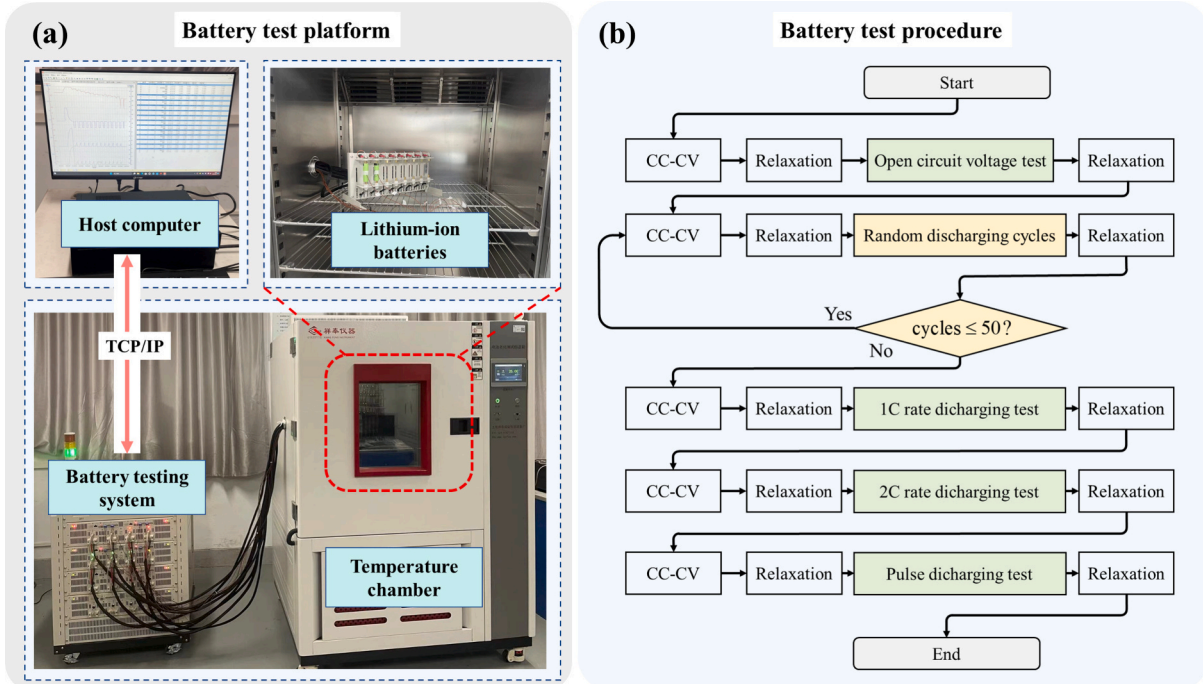
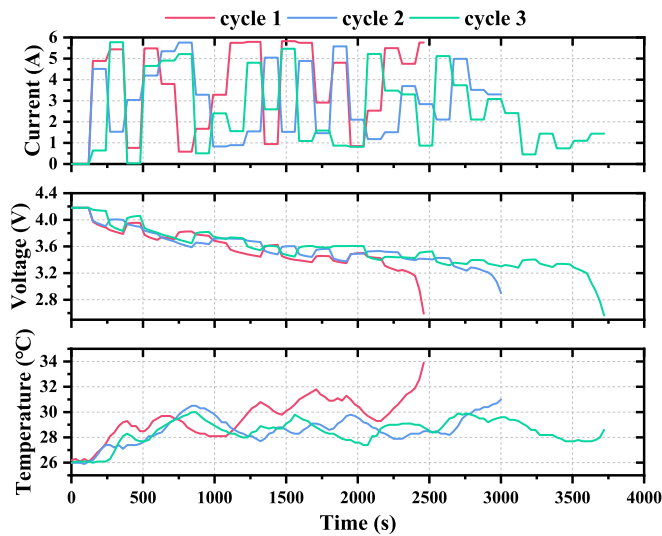


Fig. 8. LIB random discharge test (a) experimental platform and (b) procedure.

Table 2

Key parameters of three LIB datasets.

Dataset	Number	Battery type	Chemistry	Rated capacity	Rated voltage	Voltage range	Temp.	Operating condition
Experiment	1–4	DMEGC INR 18650 26E	NMC/EC-EMC/ Graphite	2.6 Ah	3.6 V	2.5–4.2 V	25 °C	Random current uniformly distributed from 0 to 6 A
Simulation	1–4	LG M50 21,700	NMC/EC-EMC/ Graphite-SiOx	5.0 Ah	3.6 V	3.0–4.2 V	25 °C	Random current uniformly distributed from 0 to 12 A
NASA(G6)	21–24	Unknown	Unknown	2.1 Ah	3.7 V	3.2–4.2 V	40 °C	Random currents with a propensity for low current levels distributed from 0.5 to 5 A
NASA(G7)	13–16	Unknown	Unknown	2.1 Ah	3.7 V	3.2–4.2 V	Ambient temp	

**Fig. 9.** Sampling values of Bat1 in the first three random discharge cycles of the experimental dataset.

in Fig. 8 was constructed. Specifically, it mainly consists of a battery testing system (Neware CT-8016-5V30A-NTA, with 16 channels), a constant temperature chamber (XF/GDW-500 L), a host computer (Windows 7.0, 4.40 GHz Intel(R) Core(TM) i5-1235U CPU), several connecting cables, Cat5e network cables, 18,650 battery test fixtures, and K-type thermocouples (ranging from -30°C to 120°C with an accuracy of $\pm 1^{\circ}\text{C}$ and a temperature resolution of 0.1°C). Four 2.6 Ah DMEGC INR 18650 LIBs were used for testing, with key parameters shown in Table 2. Additionally, an uncertainty analysis of the experimental data is provided in the Supplementary Note 1.

To ascertain a representative temperature of the batteries, K-type thermocouples were strategically placed at the midsection of the batteries to measure the surface temperature at this midpoint. A series of tests were conducted under constant temperature conditions of 25°C , including OCV tests, 1C discharging tests, 2C discharging tests, pulse discharging tests, and 50 complete random discharging cycles, as illustrated in the experimental procedure. OCV test current was set at $1/20\text{C}$, and the pulse discharging program alternated between a 10-min 1C discharging process and a 20-min rest period until reaching the cut-off voltage. The random discharge current ranged uniformly from 0 to 6 A (generated offline using MATLAB 2023a), with each step lasting 120 s before transitioning to the next step until reaching the discharge cut-off voltage. All charging processes followed the same Constant-Current-Constant-Voltage (CC-CV) charging protocol. Initially, the battery is charged at a constant current rate of $1/20\text{C}$ until the voltage reaches its upper limit of 4.2 V. Subsequently, the voltage is maintained at a constant 4.2 V while the charging continues until the charging current diminishes to below the preset cutoff current of $1/20\text{C}$, at which point the charging process is considered complete. A 30-min rest period was set between charging and discharging. Considering that the sampling

resolution of actual temperature conditions is typically greater than 0.1°C , significant data fluctuations may occur with shorter sampling periods, hence a sampling period of 30 s was chosen. Fig. 9 displays the sampled values of the first battery during the first three random discharge cycles.

3.2. Dataset 2: PyBaMM simulation data

PyBaMM is an open-source Python library used for modeling, simulating, and analyzing large-scale LIB systems [63]. It offers a rich set of features for the quick and accurate evaluation of battery system performance and behavior. In our study, we performed simulation experiments using the DFN model and lumped thermal model available in PyBaMM. The battery parameters are primarily sourced from the LG M50 21,700 cylindrical cell in literature [64,65], as detailed in Table 2. The simulation experiment settings closely mirrored those of the real experiments in Subsection 3.1, with an ambient temperature of 25°C and utilizing the same batch of random data. It is worth noting that since the simulated battery capacity is approximately double that of the experimental battery, we adjusted the current curves for the simulation experiments by multiplying the original 50 random discharge curves of each of the four batteries by 2, resulting in experimental data for four simulated batteries. Additionally, to compare the impact of different cut-off voltages, a cut-off voltage of 3.0 V was set. Fig. S4 illustrates the sampled values of the first battery during the first three random discharge cycles. Moreover, to validate the model's robustness and adaptability in responding to external environmental disturbances, an additional experiment designed around Bat1 was conducted, with details provided in Subsection 4.5.

3.3. Dataset 3: NASA random walk data

The NASA Ames Research Center supplied experimental data for twenty-eight 18,650 LIBs with a 2.1 Ah nominal capacity under seven unique operational conditions [66]. This paper analyzes eight batteries from the sixth group (operating at 40°C , denoted as Group 6 or G6) and the seventh group (conducted at room temperature, denoted as Group 7 or G7). Notably, ambient temperatures may vary during cycles conducted at room temperature. Following the same test protocol, all batteries underwent capacity and pulse assessments every 50 random driving cycles. The experiments set a cut-off voltage of 3.2 V, with stochastic currents applied in discrete values from [0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5] A, with a tendency for lower current discharges. Subjected to about 1000 random driving cycles, the batteries' available capacity fell below 60 % of their nominal value. Sampling data from the initial three random discharge cycles for batteries RW13 and RW21 are depicted in Figs. S5 and S6, respectively.

4. Results and discussion

The PIARN framework was developed using Python 3.11.4. The experimental computations were conducted on a Windows 11 64-bit system equipped with an Intel(R) Core(TM) i7-12700H CPU at 2.7 GHz and an NVIDIA GeForce RTX 3060 GPU, utilizing PyTorch 2.0.1.

This section primarily employs experimental datasets to analyze and substantiate the efficacy of the model, with further validation using simulation datasets and NASA datasets. The analysis begins with an experimental dataset to examine and corroborate the learning process and generalization capabilities of the proposed physics-informed differential block. This is followed by an evaluation of the computational efficiency and predictive accuracy of the residual attention network block. Subsequently, the performance of PIARN in forecasting across various battery types and operational environments is further validated using simulation datasets and NASA datasets. Finally, an iterative training methodology, appropriate for online deployment, is designed, and the applicability of PIARN throughout the entire life cycle of batteries is discussed using the NASA dataset.

4.1. Analysis of learning and generalization in the PIARN

In this section, we validate the learning capability of the PIARN for physical parameters using four batteries from our experimental dataset. Randomly selecting 20 discharge curves from the 50 available for each battery as the training set, the remaining curves are used as the test set. Considering that temperature sampling accuracy in practical conditions is generally greater than 0.1°C , a sampling period of 30 s is chosen to reduce fluctuations in the temperature curve. Due to the limited number of training samples, we have configured only a single batch. To enhance the model's adaptability to different environmental temperatures, we select the initial discharge temperature as the reference room temperature. Taking into account the varying lengths of discharge curves, during training, we use the maximum sequence length ($S_{\max}$) of the training samples within a batch as the sequence length (S). Other samples are adjusted to S by forward padding, with voltage (V) and temperature (T) being supplemented with the first value and current (I) with zeros. The initial learning rate is set to 0.01, and the model is trained for

a total of 500 iterations.

Initially, we conducted a sensitivity analysis and optimization of the model's hyperparameters β and D_{att} using a grid search method (see **Supplementary Note 2** for details). The analysis revealed that the performance was optimal when $\beta = 0.6$ and $D_{\text{att}} = 32$, as illustrated in **Fig. S7**. Hence, this set of hyperparameters will be chosen in the subsequent research. We then assessed the acceleration effect of the AW strategy on multi-task learning training. Focusing on the first battery, we examined the variations in voltage and temperature loss functions. **Fig. S8** illustrates the decline of different loss items during the actual training process, where some fluctuations can be observed throughout the iterations. For ease of observation, we conducted a validation using test dataset data every 50 iterations to obtain the validation results of the loss items. **Fig. 10(a)** and **(b)** depict the changes in these loss terms before and after implementing the AW. Due to the scale disparities among different tasks, the convergence of the voltage loss term was adversely impacted by the temperature loss when AW were not applied (w.o. AW), slowing overall convergence. Introducing AW (w. AW) significantly accelerated the decrease in voltage loss, with final values dropping from 0.048 V to 0.027 V. The substantial convergence of the voltage loss term illustrates that the agent model is more closely aligned with the actual battery system, enhancing the accuracy of thermal behavior simulations and temperature predictions. Consequently, the convergence value of the temperature loss term decreased from 0.253°C to 0.210°C with AW. As depicted in **Fig. 10(c)** and **(d)**, comparing test errors across four types of batteries before and after the implementation of AW shows noticeable improvements in voltage and temperature prediction accuracy, effectively validating the efficacy of AW.

Fig. 11 presents the efficacy of voltage and temperature predictions for four batteries following the incorporation of an AW mechanism, and its corresponding prediction regression plots and error histograms are shown in **Fig. S9**. The observed prediction curves for both temperature

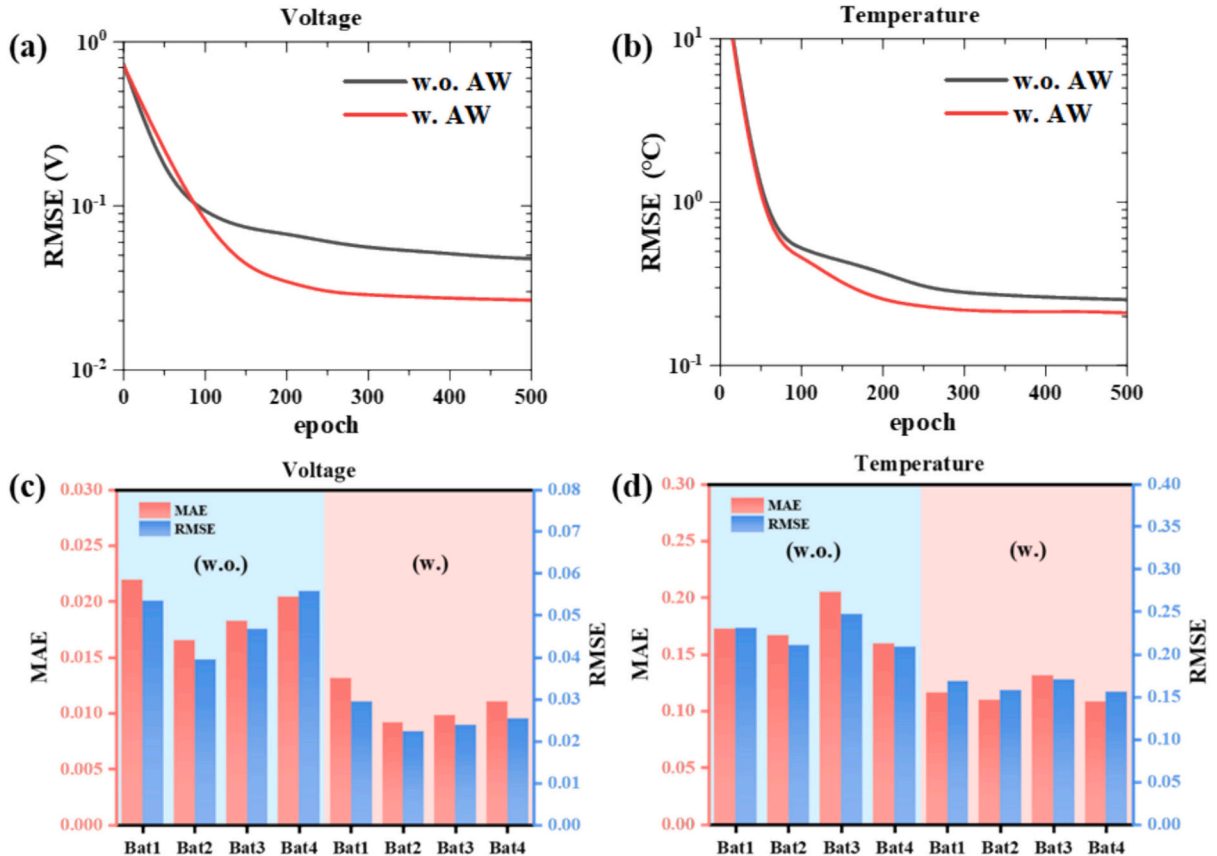


Fig. 10. Comparison of loss reduction and accuracy with and without (w./w.o.) AW (Adaptive Weighting) strategy.

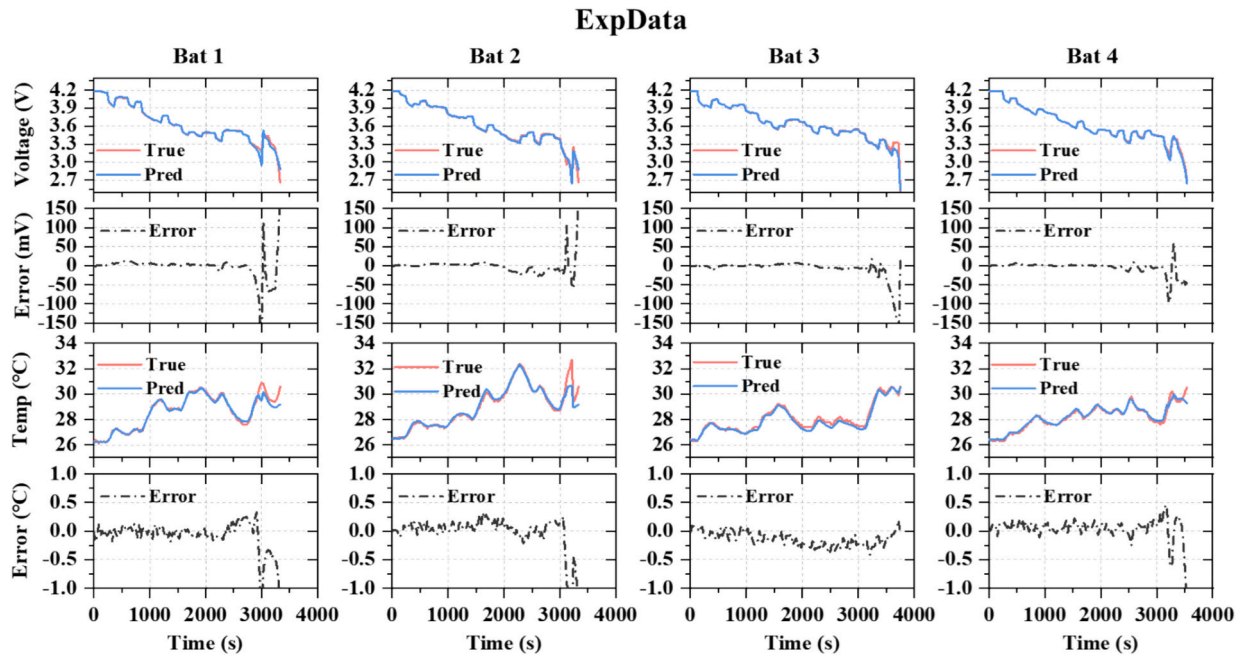


Fig. 11. Comparative analysis of voltage and temperature prediction results and associated errors for batteries in ExpData (illustrated using a single discharging sample from the test set).

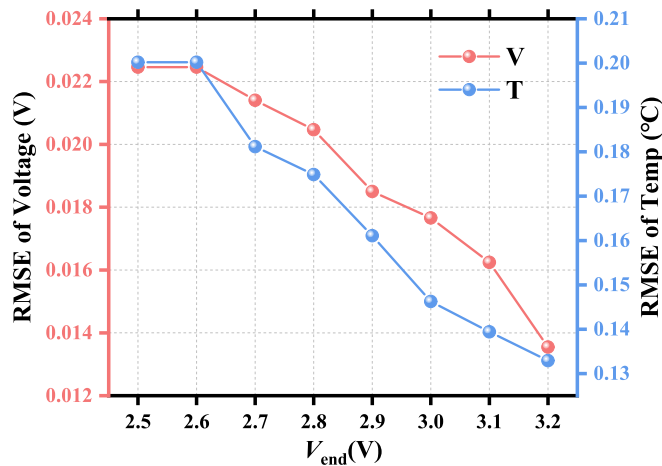


Fig. 12. Variation in prediction accuracy with cut-off voltage.

and voltage align closely with the actual curves throughout most time points, with absolute errors maintained below 20 mV and 0.3 °C respectively. During the latter section of the prediction timeline, heightened discrepancies are evident, primarily attributable to the SOC transitioning past the flat region of the OCV curve. This transition is marked by a rapid descent in OCV when SOC continues to increase (Fig. 5). At this juncture, even small deviations in coulomb counting of SOC can result in significant voltage prediction errors. Similarly, as the battery discharges to a lower SOC, internal electrochemical processes switch from reversible endothermic to exothermic reactions, prompting a steep climb in temperature that further exacerbates the prediction error [67]. In practical scenarios, users rarely drain the battery to 0 % SOC due to range anxiety; furthermore, deep discharge cycles may instigate the collapse of electrode material structures and dissolution of active substances, thereby accelerating battery degradation. As such, actual discharge depths are commonly less than the total available capacity, fortuitously bypassing the problematic low-SOC region and resulting in optimistically improved prediction outcomes. Fig. 12

illustrates the changes in prediction errors and discharge depths depending on various cutoff voltages ranging from 2.5 to 3.2 V. The results substantiate our preceding analysis, demonstrating a clear reduction in prediction errors as the cutoff voltage is raised, while the available capacity remains largely unaltered at 94 % of the capacity at 2.5 V (Fig. S10).

Furthermore, to discern the function of the physical layer within the PIARN architecture, it was juxtaposed against a suite of pure machine learning models including LSTM, GRU, and ATTGRU (Attention-GRU). Taking Bat1 as the subject, predictions for voltage and temperature were made under various discharge conditions, including different C-rates and pulse discharge scenarios, beyond the random discharge profile of the test set, as illustrated in Fig. 13. It is observed that the PIARN model not only exhibits commendable prediction acuity for random discharges but also maintains this fidelity for scenarios never encountered during training. In contrast, pure machine learning approaches incur substantial predictive errors, particularly within pulse discharge and 2C-rate discharge scenarios where the deviation from actual values is markedly pronounced. Table 3 compares the average prediction errors and relative errors of four battery types under various discharge conditions. Compared to purely machine learning-based approaches, PIARN exhibits less fluctuation in prediction errors across conditions, particularly under pulse and 2C-rate discharge scenarios. This indicates the role of the physical layer in enhancing the model's understanding of the battery's internal dynamics, thereby improving its generalization capabilities.

To verify this hypothesis, the evolution of physical parameters during training was monitored, as demonstrated in Fig. S11. The convergence of parameters to certain values after sufficient iterations suggests that the learning of physical parameters may aid in revealing the principal kinetic trends of the actual system. A noteworthy finding was that the final convergence state seemed contingent on initial parameter settings. Consequently, the learned physical parameters within the framework may represent 'true' values under specific compensatory conditions. Despite this caveat, extensive experimental observations corroborate that these learned parameters still enable the model to attain high precision and robust generalization performance.

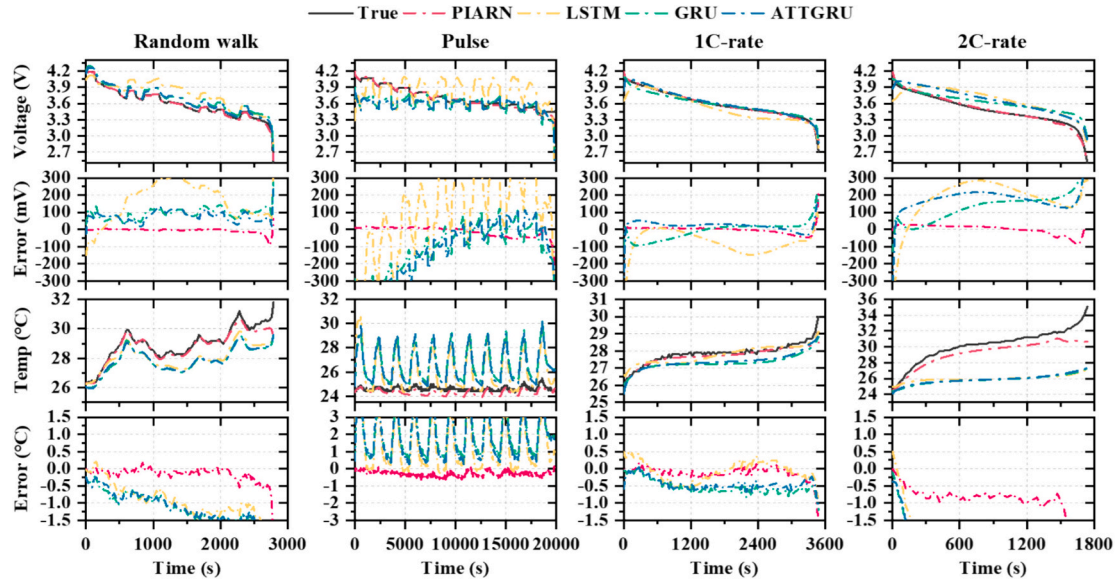


Fig. 13. Comparative analysis of voltage and temperature prediction curves, along with errors, for different discharge scenarios of Bat1 in ExpData using PIARN and other pure machine learning methods.

Table 3

Comparative analysis of voltage and temperature prediction accuracy for different discharge scenarios in experimental dataset using PIARN and other pure machine learning methods.

Variable	Methods	Random Walk		Pulse		1C-rate		2C-rate	
		RMSE	Relative	RMSE	Relative	RMSE	Relative	RMSE	Relative
Voltage	PIARN	0.027	100 %	0.046	174 %	0.027	102 %	0.035	134 %
	LSTM	0.106	100 %	0.262	248 %	0.109	103 %	0.175	166 %
	GRU	0.066	100 %	0.189	285 %	0.041	61 %	0.163	246 %
	ATTGRU	0.066	100 %	0.187	284 %	0.062	94 %	0.165	251 %
Temp	PIARN	0.318	100 %	0.270	85 %	0.270	85 %	0.961	303 %
	LSTM	0.653	100 %	2.374	363 %	0.581	89 %	4.212	645 %
	GRU	0.638	100 %	2.596	407 %	0.412	65 %	4.073	638 %
	ATTGRU	0.632	100 %	2.185	346 %	0.452	72 %	4.266	675 %

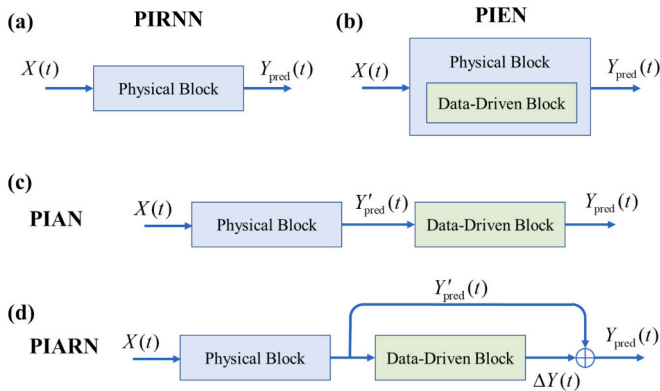


Fig. 14. Comparative illustration of neural network architectures with different compensation mechanisms.

4.2. Efficiency and accuracy analysis of residual attention networks

This subsection primarily evaluates the effectiveness of the residual attention network design through a series of comparative and ablation experiments. As mentioned earlier, the simplified NDC thermo-electrical coupled model, serving as a kernel for the physical differential block, lacks some crucial nonlinear dynamic mechanisms, such as electrochemical polarization effects associated with charge transfer at the solid-

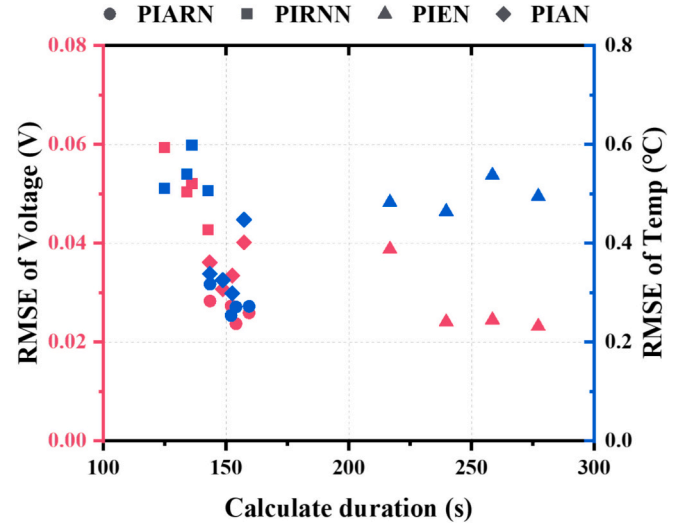


Fig. 15. Comparative analysis of prediction results with different neural network compensation mechanisms.

liquid interface and reversible entropy heat generated by electrochemical reactions. In previous studies, embedded neural networks are commonly utilized as nonlinear mapping surrogates to compensate for

these omissions, achieving commendable results in voltage prediction [44,46]. Additionally, a direct connect structure is reported in the literature [42], where the physical state variables are fed directly into a neural network for nonlinear transformation without any skip connections, yielding predictive performance comparable to residual learning strategies. However, it should be noted that the direct connect architecture discussed in the literature is based on static physical foundations, unlike the cyclic neural network framework-based physical differential block proposed in this paper. To demonstrate the advantages of the residual compensation architecture, we designed various compensation methods based on the PIRNN illustrated in Fig. 4 (here, the uncompensated pure physical network is referred to as PIRNN)—namely, the Physical Information Embedded Network (PIEN) and Physical Information Attention Network (PIAN)—and compared them against our proposed PIARN, with their structural differences displayed in Fig. 14. The specific details of PIRNN have been described in **Subsections 2.1.1 and 2.1.2**. In comparison to PIARN, PIAN eliminates skip connections to facilitate a comparison of the two connection approaches. PIEN, similar to the model presented in [44], supplements complex nonlinear mapping relationships by establishing data-driven nodes within the RNN units. The detailed description of this model can be found in the **Supplementary Note 3**.

Using random discharge data from four batteries in the experiments, we compared the training duration and prediction accuracy of these compensation methods against the purely physical approach, as depicted in Fig. 15. The results indicate that PIARN significantly reduced voltage and temperature prediction errors compared to PIRNN at a minor increase in training costs. This can be attributed to the fact that the parallelizable architecture of the PIARN's residual block can perform more efficiently compared to the time-consuming sequential computations in the PIRNN physical differential block. In contrast, although PIEN showed similar voltage prediction effectiveness to PIARN, it still demonstrated a larger temperature prediction error. This may stem from the fact that the embedded neural network increases the burden of serial computation on one hand, and on the other, promotes gradient vanishing which slows down convergence. Meanwhile, PIAN had computation times comparable to PIARN and showcased some improvement in accuracy, yet it could not match the enhancement obtained by PIARN. This suggests that the skip connections in PIARN enable the retention of more gradient information in the underlying physical model, thus improving the network's training outcomes and generalization ability.

Table 4 provides a detailed comparison of the performance across four different model architectures. It can be observed that the training time for PIEN is significantly longer than that for PIRNN, while the increase in maximum memory usage is minimal. This is primarily due to the increased depth of the RNN units, which substantially extends the training time. In contrast, the training time for PIARN shows only a slight increase compared to PIRNN, despite the maximum memory usage rising from 25.11 MB to 79.27 MB. This indirectly demonstrates that the residual attention module in PIARN can significantly enhance memory utilization efficiency and parallel computation efficiency through GPU acceleration. Furthermore, PIARN allows for the design of more network parameters, thereby improving the model's nonlinear mapping capability and prediction accuracy.

Subsequent ablation experiments were conducted to assess the

functionality of each block within the residual network. To demonstrate the superiority of physical normalization (PN), we compared it with conventional batch normalization (BN) and layer normalization (LN) in the context of PIARN—denoted as PIARN-PN, PIARN-BN, and PIARN-LN respectively, as shown in Fig. 16. In the ranking of the three normalization methods, the errors followed the order of PIARN-PN < PIARN-LN < PIARN-BN. This is because batch might not accurately represent statistical information when dealing with small training batches, and since prediction in this study is conducted on a per-sample basis, batch normalization is inapplicable. Though layer normalization relies on inter-channel or feature statistics within each sample, providing stability when handling small batches or sequential data, it still loses some information when confronted with varying distributions between different samples. Therefore, physical normalization, infused with prior knowledge, provides optimal adaptability and stability for the specific architectural characteristics of our model.

It is further observed in Fig. 16 that incorporating channel attention leads to performance surpassing models without it. Considering the complex nonlinear correlations between the residuals of voltage and temperature and the various physical quantities, we have incorporated the state information of the physical layers along with input-output information into the residual network's input. The channel attention mechanism enables the model to focus more on pertinent feature channels, mitigating the disturbance from irrelevant information. Simultaneously, as the nonlinear components and physically missing parts compensated by the residual network have inherent temporal characteristics, the introduction of the temporal block enhances the global time-dependency of features, thereby significantly improving model accuracy.

4.3. Validation on simulation dataset and NASA dataset

To further validate the potential of the proposed PIARN model in thermal management and temperature prediction, this subsection tests the model's accuracy and generalizability using both simulation dataset and the NASA dataset. Fig. 17 delineates the battery temperature profiles generated from the simulation dataset during the phases of training and prediction, alongside the evolution of error dynamics. Table 5 juxtaposes the prediction errors derived from diverse datasets, revealing the persistent robustness of the proposed approach on the simulation dataset. The implementation of an elevated cutoff voltage (3.0 V) is correlated with enhanced preciseness in the simulation data, as indicated by impressively low RMSE of 0.1197, respectively. The battery parameters and environmental conditions within the simulated environment are highly regulated, circumventing measurement inaccuracies and the accumulation of SOC errors. Consequently, even when juxtaposed with ExpData (3.0 V), which shares an identical cutoff voltage, the simulation dataset maintains markedly lower total errors. Fig. S12 illustrates the variations in prediction error across different ambient temperatures, demonstrating the PIARN model's robust adaptability under various temperature conditions. Moreover, we conducted a comparison of the prediction errors under different operational conditions after training with random-walk operating conditions from the simulation dataset, as explicated in Table 6. The outcomes validate the formidable generalization capabilities of the PIARN methodology within the simulation environment.

Fig. 18 and Fig. 19 present the expected battery temperature curves under different ambient temperatures derived from the NASA dataset, along with the prediction errors. Combined with Table 5, it can be observed that the temperature prediction error slightly exceeds the error of the previous dataset. The sensitivity of temperature to ambient conditions might elucidate this condition. For instance, in NASADData (G7), conducted at an open room temperature, potential fluctuations in room temperature and air currents could emerge. At a temperature of 40 °C in NASADData (G6), the prediction accuracy slightly decreases, which might be due to the inherent differences in the OCV curves at different

Table 4
Comparison of model performance across four physical information neural network architectures.

Model	Average Training Time (s)	Maximum Memory Usage (MB)	Average Voltage RMSE (V)	Average Temperature RMSE (°C)
PIRNN	134.44	25.11	0.0511	0.5384
PIEN	248.05	27.84	0.0277	0.4949
PIAN	150.41	79.04	0.0351	0.3525
PIARN	152.22	79.27	0.0263	0.2786

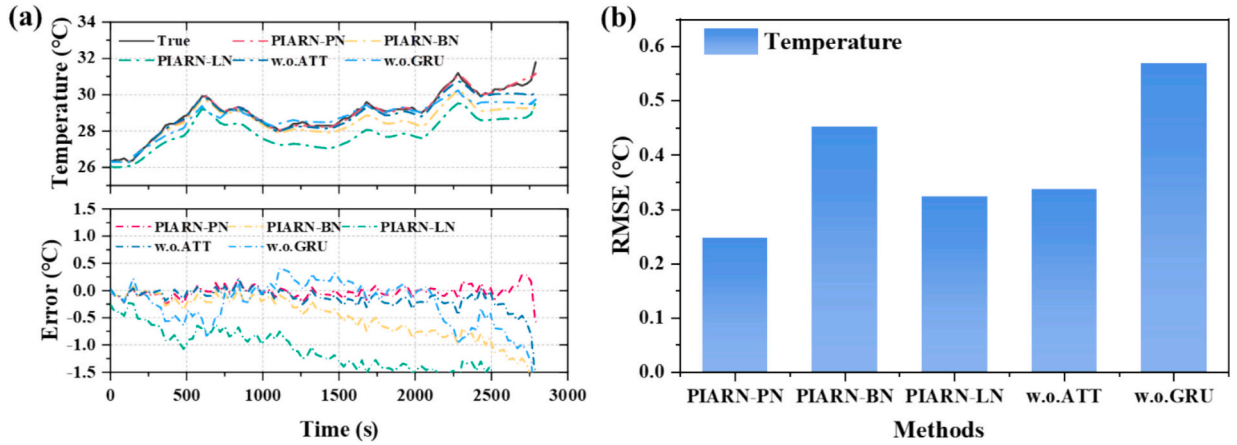


Fig. 16. Empirical study on the ablation analysis of residual network architecture in PIARN.

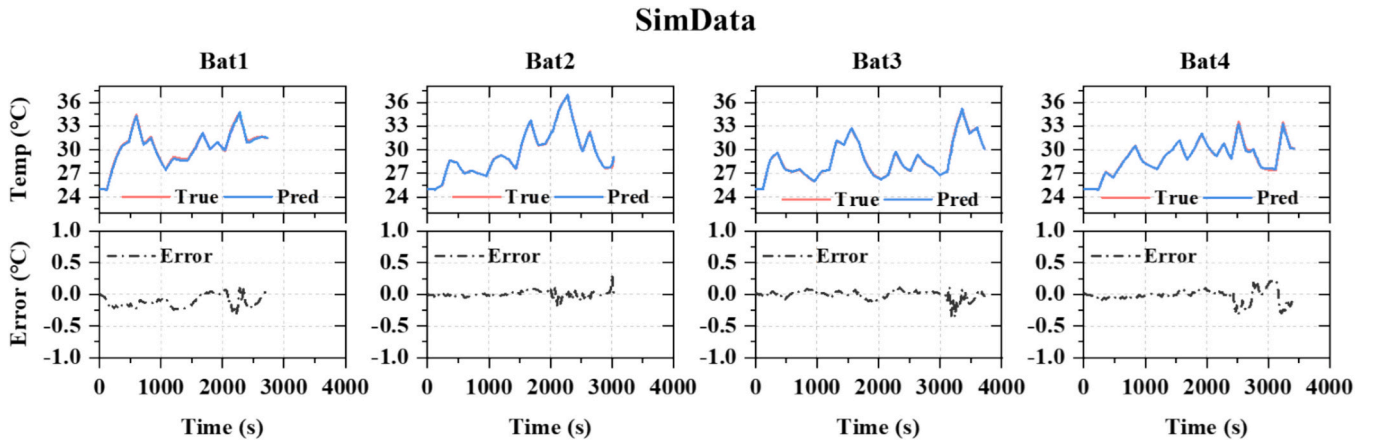


Fig. 17. Comparative analysis of temperature prediction results and associated errors for batteries in simulation dataset (illustrated using a single discharging sample from the test set).

Table 5
Comparative analysis of temperature prediction errors across different datasets (Unit: °C).

Battery	Metrics	ExpData(2.5 V)	ExpData(3.0 V)	SimData	NASADData(G6)	NASADData(G7)
Bat1	MAE	0.1493	0.1316	0.1138	0.7010	0.2611
	RMSE	0.2564	0.1679	0.1446	0.8661	0.3020
	R ²	0.9457	0.9642	0.9954	0.8426	0.9819
Bat2	MAE	0.1574	0.1345	0.0692	0.5227	0.1773
	RMSE	0.2231	0.1792	0.0938	0.6296	0.2186
	R ²	0.9461	0.9617	0.9969	0.8950	0.9887
Bat3	MAE	0.1701	0.1626	0.0652	0.6774	0.2852
	RMSE	0.2428	0.1955	0.0900	0.7872	0.3419
	R ²	0.9301	0.9397	0.9982	0.8180	0.9730
Bat4	MAE	0.1454	0.1125	0.1083	0.3460	0.2559
	RMSE	0.2708	0.1462	0.1505	0.4015	0.2846
	R ²	0.9552	0.9786	0.9977	0.9358	0.9801
Total	MAE	0.1555	0.1353	0.0891	0.5618	0.2449
	RMSE	0.2483	0.1722	0.1197	0.6711	0.2868
	R ²	0.9443	0.9610	0.9971	0.8728	0.9809

temperatures and the typical increase in airflow when the temperature rises. Despite these factors, the robust nonlinear fitting prowess of the residual neural network mitigates temperature discrepancies, ensuring coherence between predicted and actual trends across multiple contexts. Thus, the PIARN model not only demonstrates high precision and strong generalizability but also maintains formidable adaptability and robustness.

Fig. 20 compares the prediction errors of the PIARN model trained on

training sets of different sizes. It was observed that even with a small-scale sample containing only 5 training curves, the model achieved good predictive performance when incorporating prior knowledge of physics.

Notably, when applying the PIARN model to datasets with different battery configurations, it requires only a single offline measurement to obtain the OCV-SOC relationship curve and the rated capacity for that specific battery type. Other physical parameters do not require precise

Table 6

Comparative analysis of temperature prediction errors (RMSE) for Bat1 in simulated dataset across various operating conditions (Unit: °C).

Battery	Random Walk	Pulse	1C-rate	2C-rate
Bat1	0.1445	0.3484	0.1333	0.2725
Bat2	0.0921	0.3834	0.0557	0.4612
Bat3	0.0964	0.6423	0.0952	0.4159
Bat4	0.1505	0.4251	0.1318	0.3827
Total	0.1209	0.4498	0.1040	0.3831

offline identification and can be shared across different battery types, as illustrated in **Table S1**. This approach significantly reduces the need for offline parameter identification and holds promise for cost-effective scalability across a wider range of battery types, such as those with varying chemical compositions and capacities.

4.4. Iterative updating of models for the full LIB lifecycle

This subsection delves into the viability of implementing PIARN in real-world vehicular systems. Prior studies demonstrated the high accuracy and robustness of PIARN for predicting temperature during the stochastic discharge of batteries, maintaining good predictive performance even with limited training samples. Thus, PIARN shows promising application prospects for predicting temperature throughout the entire lifecycle of LIBs in vehicular or energy storage systems. However, due to its serial physical underpinnings, this method requires a significant computation cycle even on experimental computers. When

considering the limited computational efficiency of onboard BMS, applying PIARN for voltage and temperature prediction in such systems necessitates extra computational resources and the design of more effective operational strategies.

As depicted in **Fig. 1**, future vehicular BMS could leverage cloud computing or onboard Artificial Intelligence (AI) chips for enhanced computational resources, thanks to advances in communication technologies and GPU computation [68,69]. To reduce computational costs, inspired by transfer learning techniques, we devised a low-cost online iterative training strategy shown in **Fig. 21**. Unlike the random selection of training samples from adjacent discharge curves, recent history window data is used for training when applied online (denoted by Batch size), predicting subsequent voltage and temperature for the forecast window (Horizen). Over time, new training and prediction windows are selected, continually updating the model. To avoid excessive computational costs due to repeated training, training is based on the previous model iteration, decreasing the number of iterations per training session. Specifically, taking the first battery from NASADData (G7) as an example, we selected random discharge curves from the initial 500 cycles (with usable capacity decreasing from 100 % to 70 % rated capacity), setting both history and prediction window lengths to 5. This ensures small training samples while preventing model obsolescence due to battery aging or environmental changes. The initial PIARN battery model was established using discharge data from the first 5 cycles as the training set, undergoing 50 iterations, then predicting voltage and temperature for the next 5 cycles (cycles 6 to 10). Subsequently, at the second time node, new training windows are selected, and further

NASADData(G6)

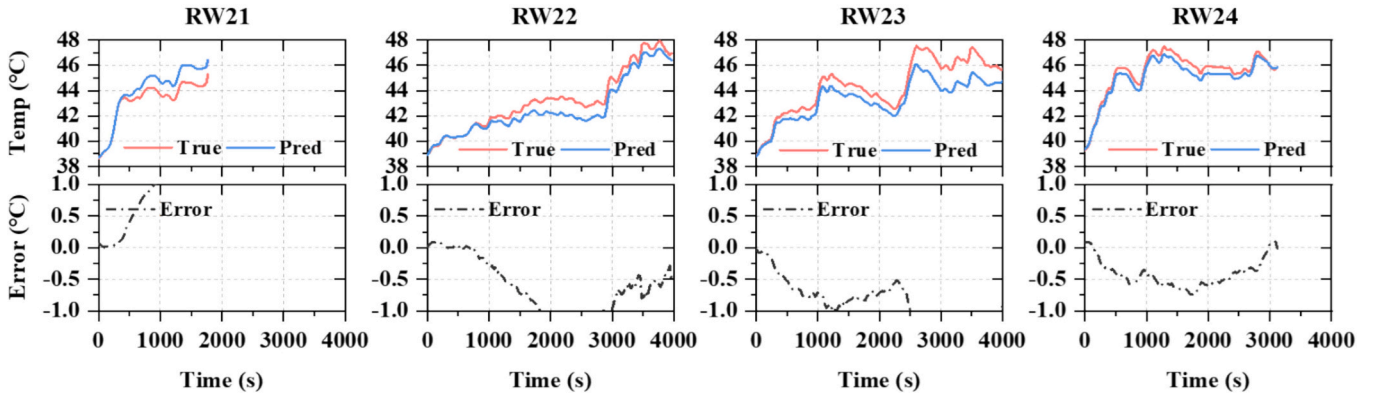


Fig. 18. Comparative analysis of temperature prediction results and associated errors for batteries in NASADData(G6) (illustrated using a single discharging sample from the test set).

NASADData(G7)

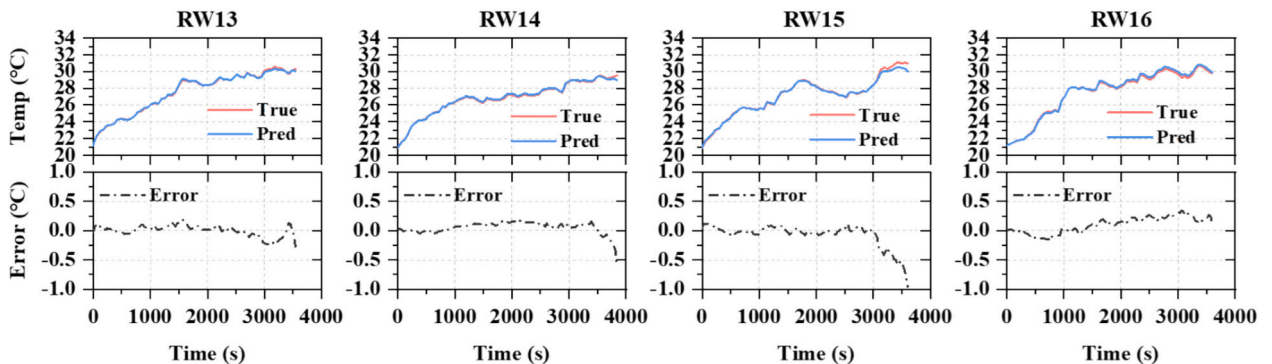


Fig. 19. Comparative analysis of temperature prediction results and associated errors for batteries in NASADData(G7) (illustrated using a single discharging sample from the test set).

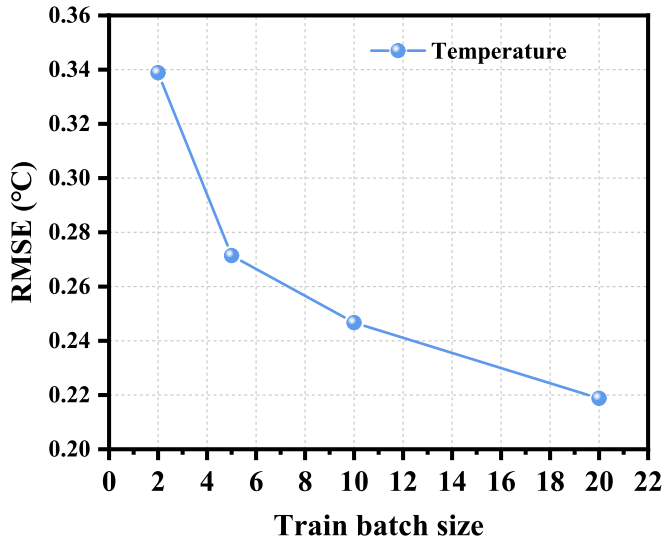


Fig. 20. Comparative assessment of the PIARN model's predictive outcomes post-training with varied training set sizes.

training is performed upon the pre-trained model, again with 50 iterations. As later trainings are fine-tuned, smaller learning rates can progressively be set. The learning rate for the n -th training step is defined by the formula:

$$l = \frac{l_0}{\left(\frac{n}{2h} + 1\right)} \quad (37)$$

Here, l_0 represents the initial learning rate set at 0.01 for this research, and h is the prediction horizon.

Fig. 22(a) depict the outcomes of the learning process for NASADData (G7), whereas Fig. 22(b) illuminates the trend in the mean prediction accuracy across individual cells as a function of cycling iterations. Observations indicate that during the initial forecast at cycle 10 (cyc10), a degree of error was present due to a limited amount of training iterations. As training continued with repeated iterations, there was a progressive enhancement in model prediction accuracy. Consequentially, the RMSE for voltage declined to approximately 0.01 and for temperature to roughly 0.1. This precision nearly surpasses that which is typically achieved with conventional training methodologies. To enhance early prediction accuracy, increasing the number of iterations in the initial training phase, such as to 500, could achieve more thorough convergence. Fig. 22(b) shows that temperature RMSE decreases within the first 120 cycles. Subsequently, the errors from both training methods stabilize and align. The choice between training strategies depends on their intended application. Overall, both strategies effectively track battery aging while ensuring high accuracy. Notably, batteries produced in modern automated factories exhibit high consistency. Thus, differing batteries can share initial models to some extent, with manufacturers potentially pre-setting these models. This approach reduces the average computational resources required for initial training, facilitating the practical application of these training methods.

4.5. Online thermal warning for the full life cycle of the LIB

To achieve optimal performance, LIBs must operate within a specific temperature range (15–35 °C) [70]. Operating outside this range can degrade the battery's performance, lifespan, and safety [13]. Accurate temperature prediction and appropriate thermal warning threshold settings are crucial for optimizing thermal management system operation, aiding drivers in making informed decisions, and enhancing the safety features of EVs (as shown in Fig. 1). Subsection 4.4 demonstrates that the presented PIARN model achieves accurate lifetime battery

temperature predictions through an online iterative training strategy, facilitating precise thermal warnings. In this subsection, we set the temperature warning threshold at 35 °C and explore the feasibility of applying this approach for temperature warnings in real-world conditions.

Fig. 23 illustrates the thermal warning effects achieved through a whole-life cycle online iterative training strategy for Bat1 from NASADData(G7). Fig. 23(a) compares predicted temperatures to actual measurements at the 20th cycle. The white region below 35 °C represents the normal operating area, while the red region above this threshold indicates the thermal warning zone. The close alignment between the predicted and actual temperatures suggests that predictions can effectively preempt temperature warnings, thereby mitigating the risk of high temperatures. To assess the accuracy of these thermal warnings, we introduce the thermal warning correctness rate, calculated as follows [71]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (38)$$

Here, TP (True Positives) denotes the count of correctly predicted normal operating instances, TN (True Negatives) the count of accurately predicted thermal warning instances, FP (False Positives) the count of incorrectly predicted as normal operating, and FN (False Negatives) the count of incorrectly predicted as thermal warning instances.

Fig. 23(b) presents the temperature prediction error and thermal warning accuracy throughout a full lifecycle iterative process using the PIARN model. Notably, the thermal warning accuracy consistently exceeds 98 %, stabilizing at 100 % as iterations progress and prediction errors diminish. Initial cycles exhibit fluctuating accuracy rates, highlighting the dynamic adaptation of the model. Importantly, Bat1 from NASADData(G7) operates under open room conditions, subject to variations in battery aging, ambient temperature, and cooling conditions similar to real-world settings. This underscores the feasibility of employing PIARN as an operational proxy model for thermal warnings.

To further validate the adaptability of our approach under changing environmental conditions, we conducted 200 dynamic discharge cycles for a single battery via PyBaMM, adhering to dynamic protocols as outlined in Subsection 3.2 (SimData_Bat1). However, environmental conditions were altered every 50 cycles. In this study, we altered the environmental conditions by controlling the heat transfer coefficient h_{th} (which represents the convective heat transfer performance between the battery surface and the airflow). This approach simulates the variations in surface airflow and heat dissipation rates that a battery experiences in different real-world scenarios. Initially, as depicted in Fig. 24(a), the ambient temperature T_a was set at 25 °C and heat transfer coefficient h_{th} at 10 W·m⁻²·K⁻¹. At the completion of the 50th cycle, T_a increased to 30 °C. By the 100th cycle, h_{th} was adjusted to 10 W·m⁻²·K⁻¹. Finally, at the 150th cycle, T_a was reduced back to 25 °C. Different environmental conditions are distinguished by varying background colors in Fig. 24(a). This dataset exclusively considers changes in environmental conditions, excluding battery aging effects.

Fig. 24(b) displays the variation in temperature prediction error and thermal warning accuracy throughout the lifecycle via iterative online training using the PIARN model. Early cycles show a gradual decrease in temperature prediction errors with thermal warning accuracy approaching 100 %, akin to empirical results from the NASADData. The PIARN model incrementally “learns” the electrothermal characteristics of the battery under current environmental constraints, enabling precise temperature predictions. Temperature prediction errors initially increase with each environmental shift, momentarily reducing thermal warning accuracy. However, as iterations progress and the model trains with new operational data, prediction errors swiftly decrease and thermal warning accuracy approaches nearly 100 % again.

Thus, the PIARN agent model adeptly tracks battery operational states and adapts to environmental changes through online iterative training, enhancing its precision and robustness in temperature

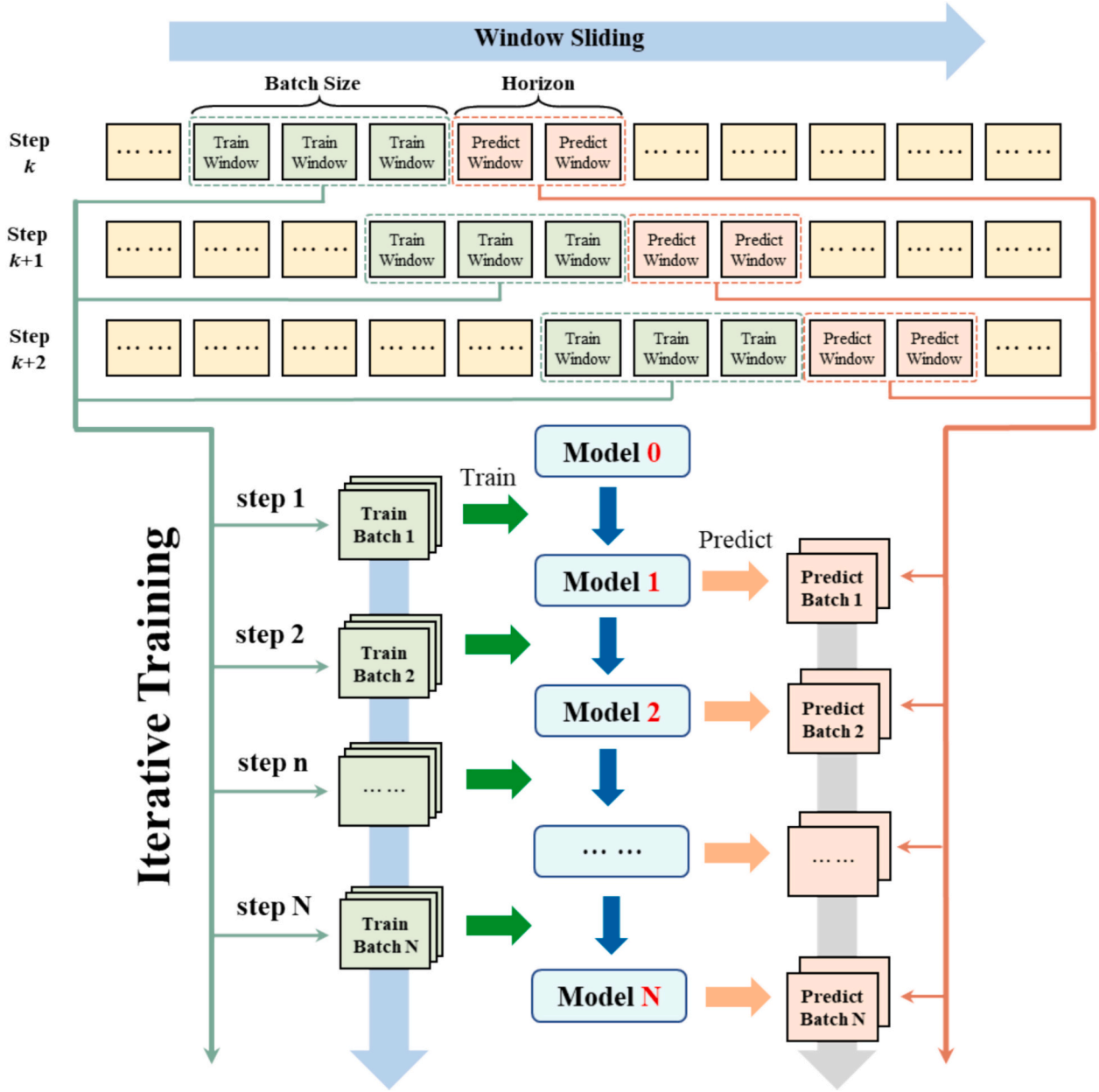


Fig. 21. Iterative online training architecture for Li-ion battery full lifecycle application.

prediction and thermal warning for real-world EV energy storage systems.

4.6. Future work

The preceding discussion has validated the PIARN model's distinct advantages over traditional architectures like PIRNN and PIEN in terms of interpretability, computational efficiency, and generalization performance. Additionally, the proposed online iterative training strategy plays a crucial role in enhancing the model's adaptability to battery aging and environmental temperature variations. However, there are still several potential areas that warrant further exploration and discussion in the future.

First, applicability under extreme conditions. Although PIARN achieves good generalization capabilities through its physical foundation, allowing it to anticipate unknown conditions within certain limits, there remains a possibility of failure under extreme conditions (e.g., higher C-rate currents, extreme environmental temperatures). Future research

will delve deeper into this issue.

Second, overcompensation in residual neural networks. While it has been demonstrated that the introduction of residual neural networks and the PIARN architecture significantly improve prediction accuracy, there is still a tendency for overcompensation, which can affect the learning of physical parameters and, consequently, the model's interpretability and generalization. Future studies should focus on optimizing gradient distribution within the physical model to more effectively guide network convergence.

Third, optimization of the computational framework and further enhancement of efficiency. Although this study has improved parallel computation efficiency through the simplification of the physical foundation and the design of the residual model, the serial nature of the physical foundation remains a major constraint on computational efficiency in long sequence prediction problems. Future work will explore the potential of applying different computational frameworks and physical models to achieve further improvements in computational efficiency.

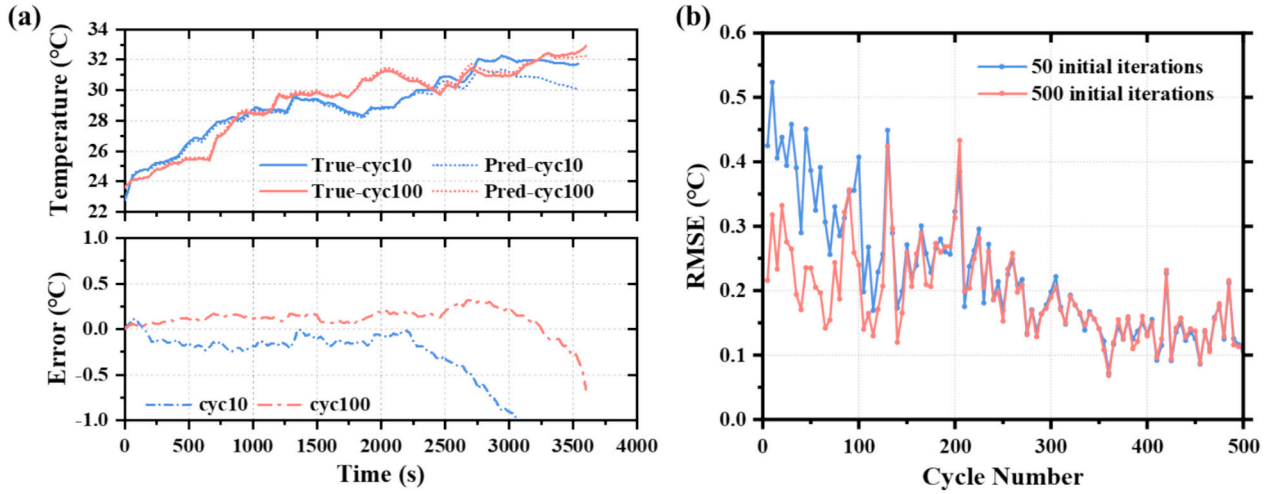


Fig. 22. Predictive results of voltage and temperature in Li-ion battery full lifecycle online iterative training strategy.

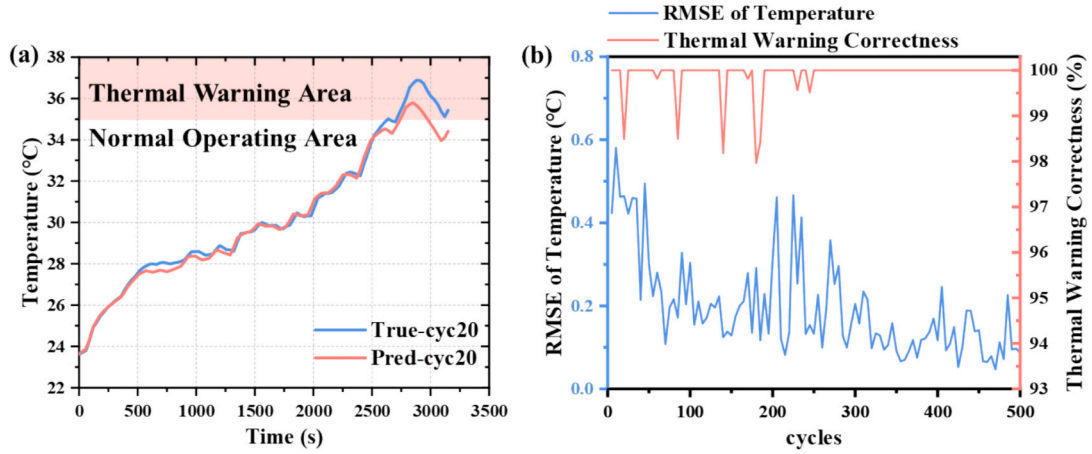


Fig. 23. Online thermal warnings for the full life cycle of Bat1 in the NASADData(G7).

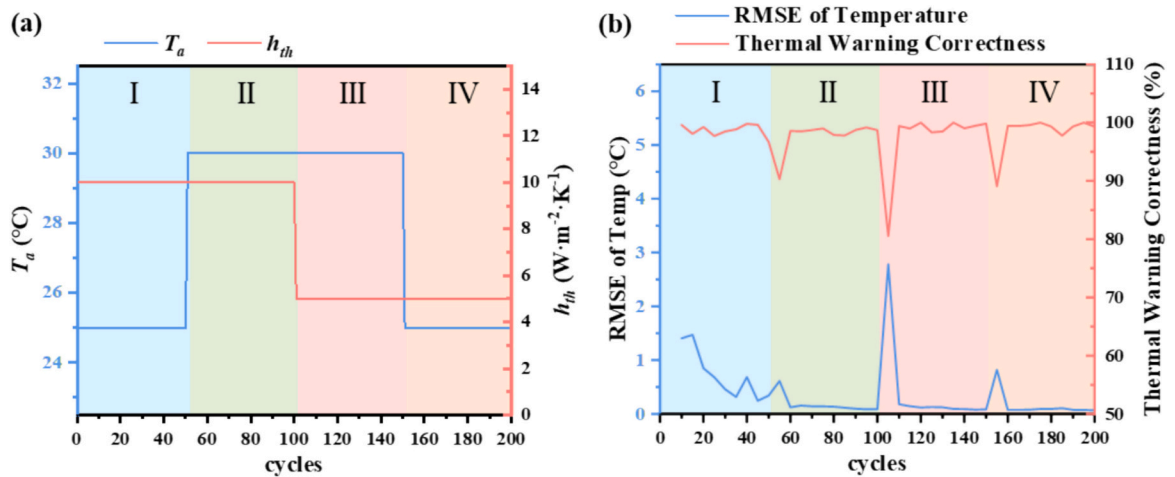


Fig. 24. Thermal warning robustness testing of SimData battery subjected to thermal environment changes.

Fourth, expansion of applications to more battery systems. A significant advantage of PIARN is its ability to learn physical parameters through backpropagation, without relying on precise offline parameter identification beyond the SOC-OCV relationship and rated capacity. This enables cost-effective scalability across different battery types. This

paper has already demonstrated PIARN's excellent scalability and versatility across various datasets. Future research will explore its application expansion to more battery systems, such as next-generation solid-state battery systems.

5. Conclusion

To enhance thermal safety management in EV LIBs, this report introduces a novel PIARN to address bottlenecks and challenges in accuracy and efficiency concerning electrothermal coupled modeling of LIBs. The framework of PIARN encompasses a physical information differential block and residual neural network block. Enhanced performance of PIARN has been realized through a synergistic integration of physical modeling with data-driven deep fusion techniques, thereby exhibiting significant advantages in prediction accuracy, generalization ability, and computational efficiency. Case studies from multiple datasets permit us to draw the following conclusions:

- (1) The AW approach with dynamic decay factors enhances voltage prediction precision by balancing optimization tasks and accelerating network convergence. The modified NDC model captures essential system dynamics, offering better generalization than traditional machine learning models.
- (2) Within the dynamic PIARN framework, the introduction of learning for “missing physics” via the residual attention neural network effectively elevates model precision. Compared to other hybrid compensation methods, the residual attention neural network augments parallel computational efficiency and training effectiveness. Analysis of diverse datasets shows that PIARN achieves high predictive accuracy and generalization with fewer samples across various voltage levels and temperatures.
- (3) The iterative updating strategy proposed for LIB life-cycle models facilitates highly accurate temperature predictions and thermal warnings at reduced training costs while concurrently monitoring the battery’s aging state. After multiple iterations, this strategy yields a temperature prediction RMSE of just 0.1 °C and achieves nearly 100 % accuracy in thermal warnings.

CRedit authorship contribution statement

Xue Ke: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lei Wang:** Visualization, Methodology, Formal analysis. **Jun Wang:** Visualization, Methodology, Conceptualization. **Anyang Wang:** Methodology, Formal analysis. **Ruilin Wang:** Resources. **Peng Liu:** Formal analysis. **Li Li:** Visualization. **Rong Han:** Visualization. **Yiheng Yin:** Methodology. **Feng Ryan Wang:** Resources, Conceptualization. **Chun-guang Kuai:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Yuzheng Guo:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2025.125627>.

Data availability

Data will be made available on request.

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